

The PCP Theorem

and its consequences for the approximation of hard problems

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IV Weekend Ortogonale – Procida, 16 Maggio 2026



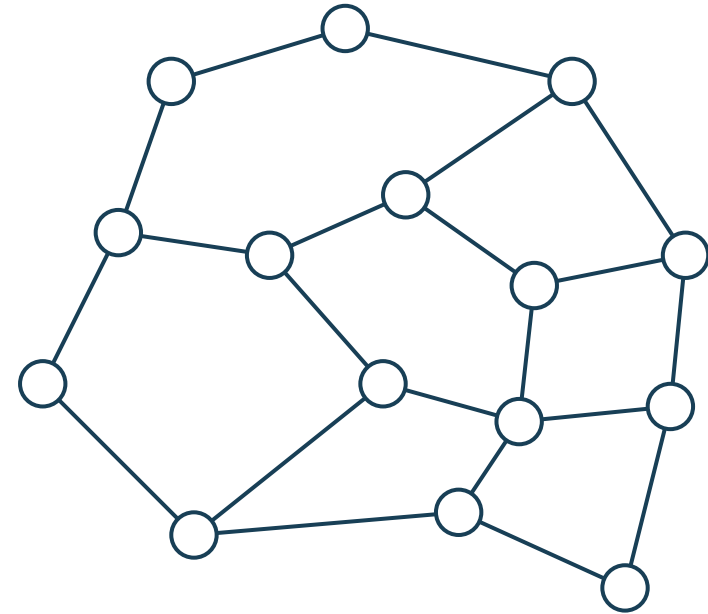
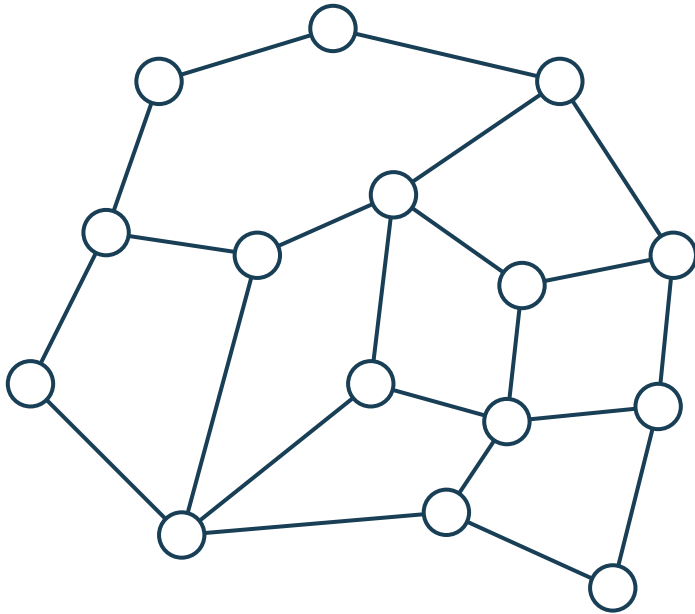
1. **P vs NP**
2. The PCP Theorem
3. Inapproximability of Max-Clique
4. Landscape

P: 2-Colorability

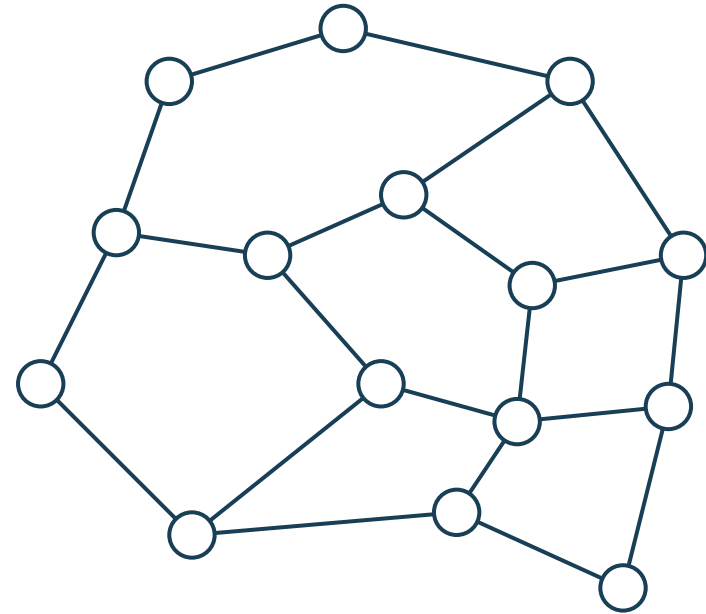
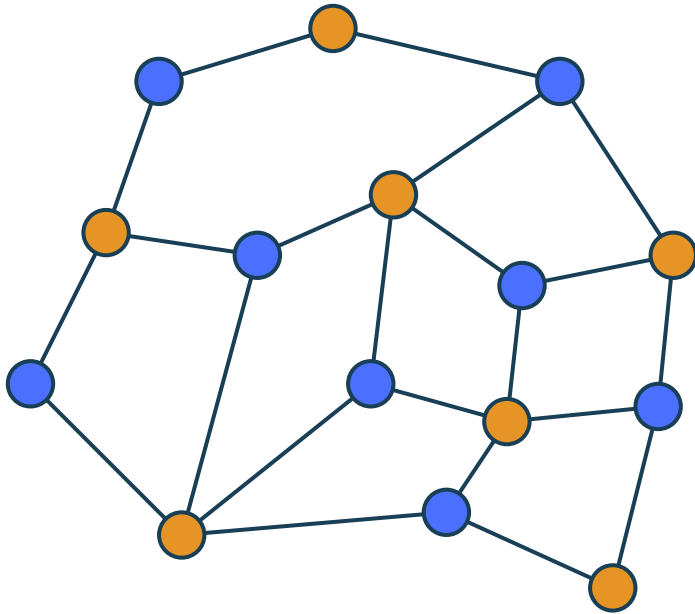
Problem (2-Colorability)

Given a graph $G(V, E)$, decide whether it is 2-colorable.

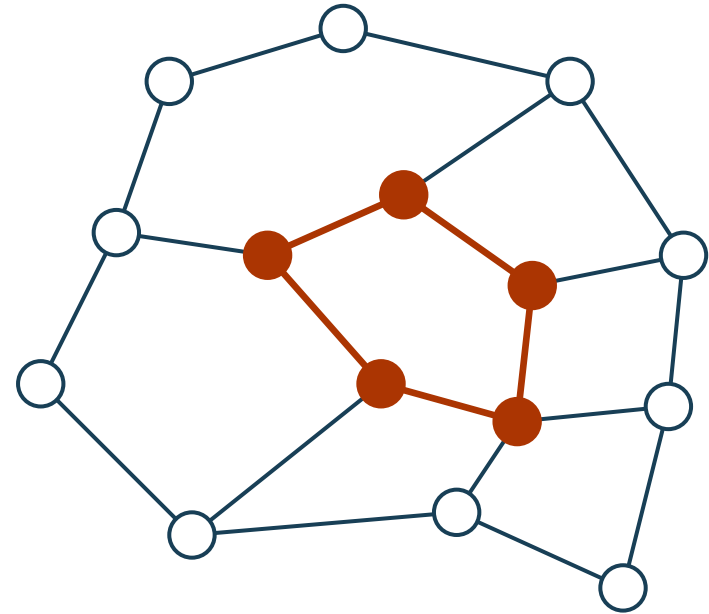
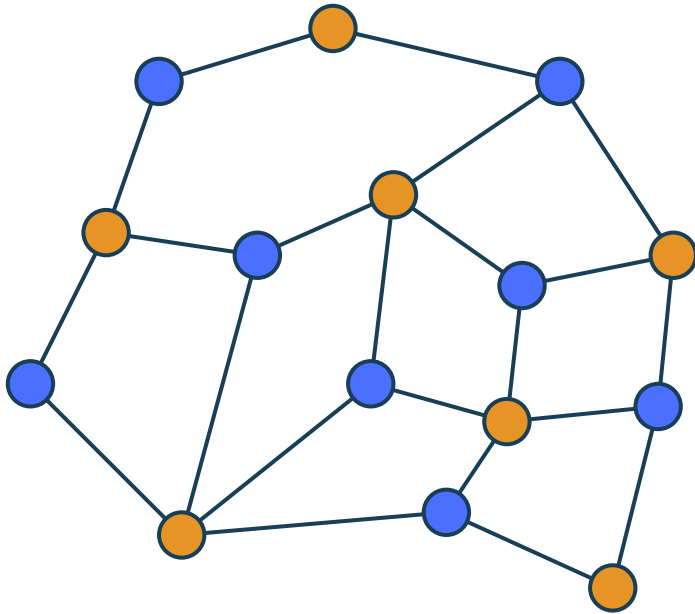
P: 2-Colorability



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What does *efficient* mean?

efficient $\Leftrightarrow$ polynomial time.

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2-Colorability can be solved efficiently in $O(|V| + |E|)$.

NP: 3-Colorability

Problem (3-Colorability)

Given a graph $G(V, E)$, determine whether it is 3-colorable.

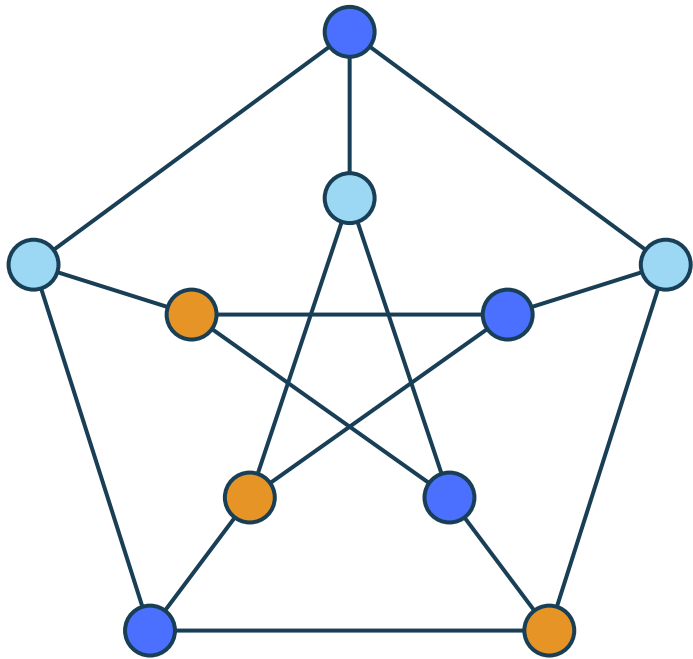
NP: 3-Colorability

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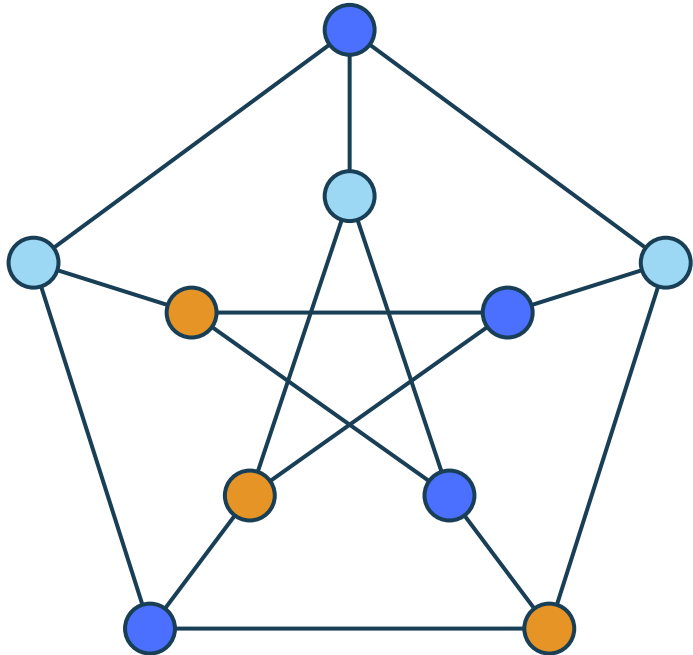
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DRASTIC CHANGE!

NP: 3-Colorability

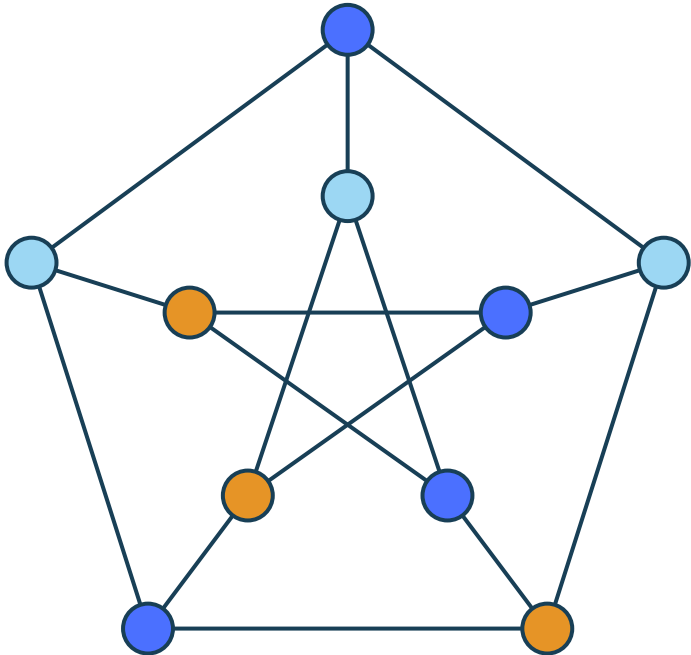


NP: 3-Colorability



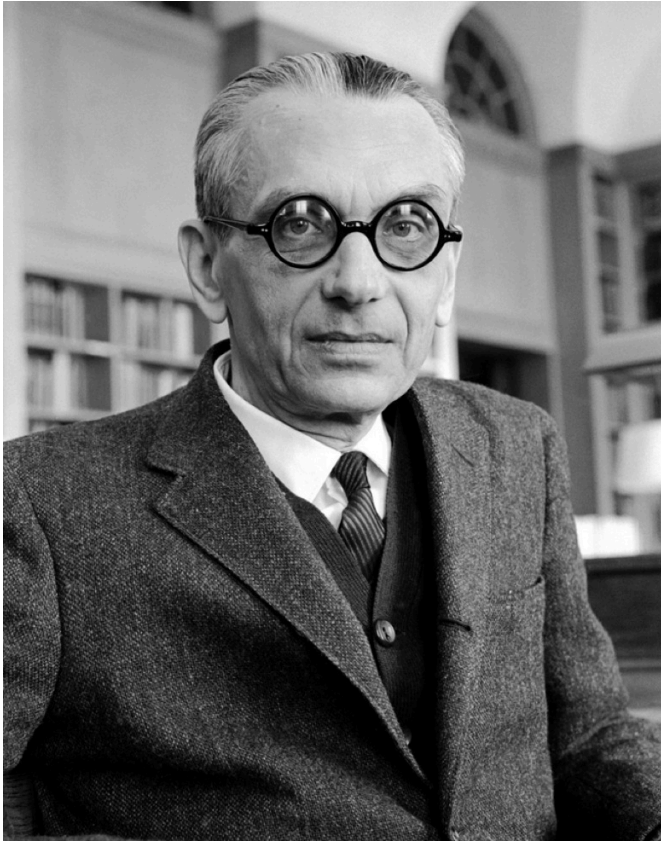
- No efficient algorithm is known.

NP: 3-Colorability



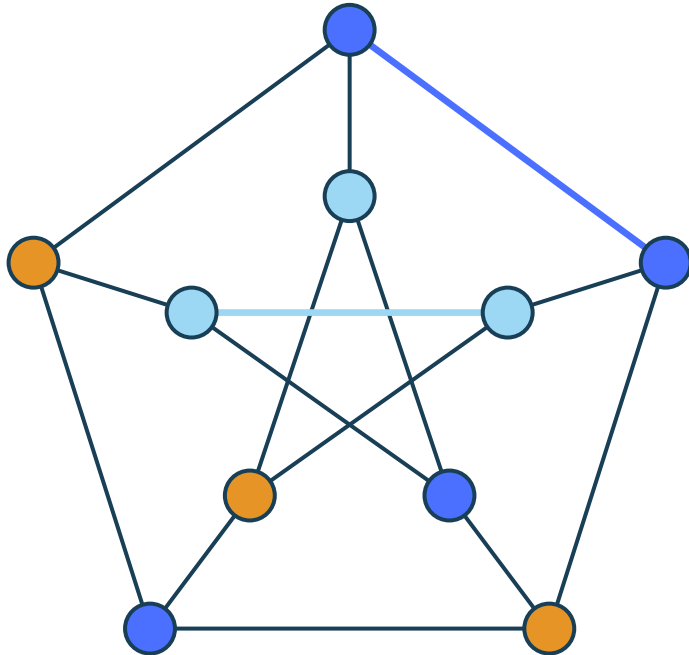
- No efficient algorithm is known.
- It is conjectured it doesn't exist (P vs NP).

NP: 3-Colorability



- No efficient algorithm is known.
- It is conjectured it doesn't exist (P vs NP).
- If it did, mathematical proofs could be *found* efficiently.

What can we say about 3-Colorability?



Correctness can be
efficiently **verified**.

P vs NP

P

Decision problems
whose solution
can be **found** efficiently.

NP

Decision problems
whose solution
can be **verified** efficiently.

Millenium Problem

P vs NP

Is P a proper subset of NP?

Millenium Problem

P vs NP

Is P a proper subset of NP?

Is it true that 3-Colorability and thousands of other problems **cannot** be solved efficiently?

Why care about 3-Colorability?

Karp (1972); Lovász (1973)

Every problem in NP can be formulated in terms of 3-Colorability.

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10/39

Probabilistically checkable proofs

New Short Cut Found For Long Math Proofs

A proof can be tested by checking just a part, inventors say.

By GINA KOLATA

IN a discovery that overturns centuries of mathematical tradition, a group of graduate students and young researchers has discovered a way to check even the longest and most complicated proof by scrutinizing it in just a few spots.

The finding, which some mathematicians say seems almost magical, depends upon transforming the set of logical statements that constitute a proof into a special mathematical form in which any error is so amplified as to be easily detectable.

Using this new result, the researchers have already made a landmark discovery in computer science. They showed that it is impossible to compute even approximate solutions for a large group of practical problems that have long foiled researchers. Even that negative finding is very significant, experts say, because in mathematics, a negative result,

showing something is impossible, can be just as important and open just as many new areas of research as a positive one.

The discovery was made by Sanjeev Arora and Madhu Sudan, graduate students at the University of California at Berkeley, Dr. Rajeev Motwani, an assistant professor at Stanford University, and Dr. Carsten Lund and Dr. Mario Szegedy, young computer scientists at A.T.&T. Bell Laboratories. Dr. Motwani, who is the senior member of the group, just turned 30 on March 26.

"With the conventional notion of a proof, you had to check it line by line," said Dr. Michael Sipser, a theoretical computer scientist at the Massachusetts Institute of Technology. "An error might be buried on page 475, line 6. A 'less than or equal to' should have been a 'less than.' That would totally trash the whole proof. But you'd have to dig through the whole thing to find it," Dr. Sipser said. Now, he added, "the new idea is that there is a way to transform any proof so that if there is an error, it appears almost everywhere. I'd say, 'You have a proof? Show me a page.' If there is an error, it will be there."

The finding, which is built on two and a one half years work by leading

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The PCP Theorem implies this is possible.

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The class $\text{PCP}(O(\log n), O(1))$

Let $c, q \in \mathbb{Z}^+$ be constants and V be an efficient probabilistic verifier. On input $x \in \{0, 1\}^n$, to verify a proof $\Pi : \mathbb{Z}^+ \rightarrow \{0, 1\}$:

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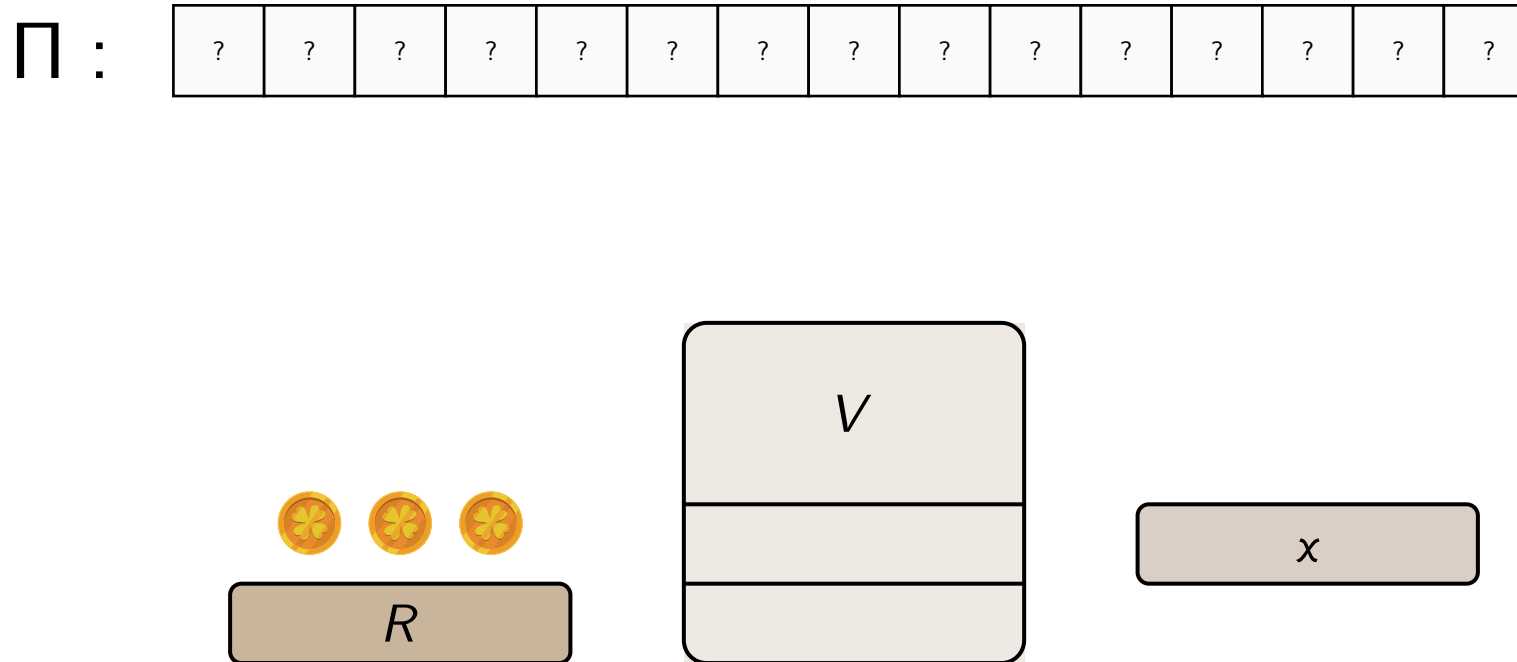
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- based on R and $\Pi(i_1), \dots, \Pi(i_q)$, V decides whether to accept or reject.

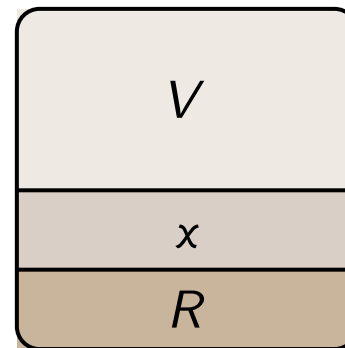
The PCP Theorem

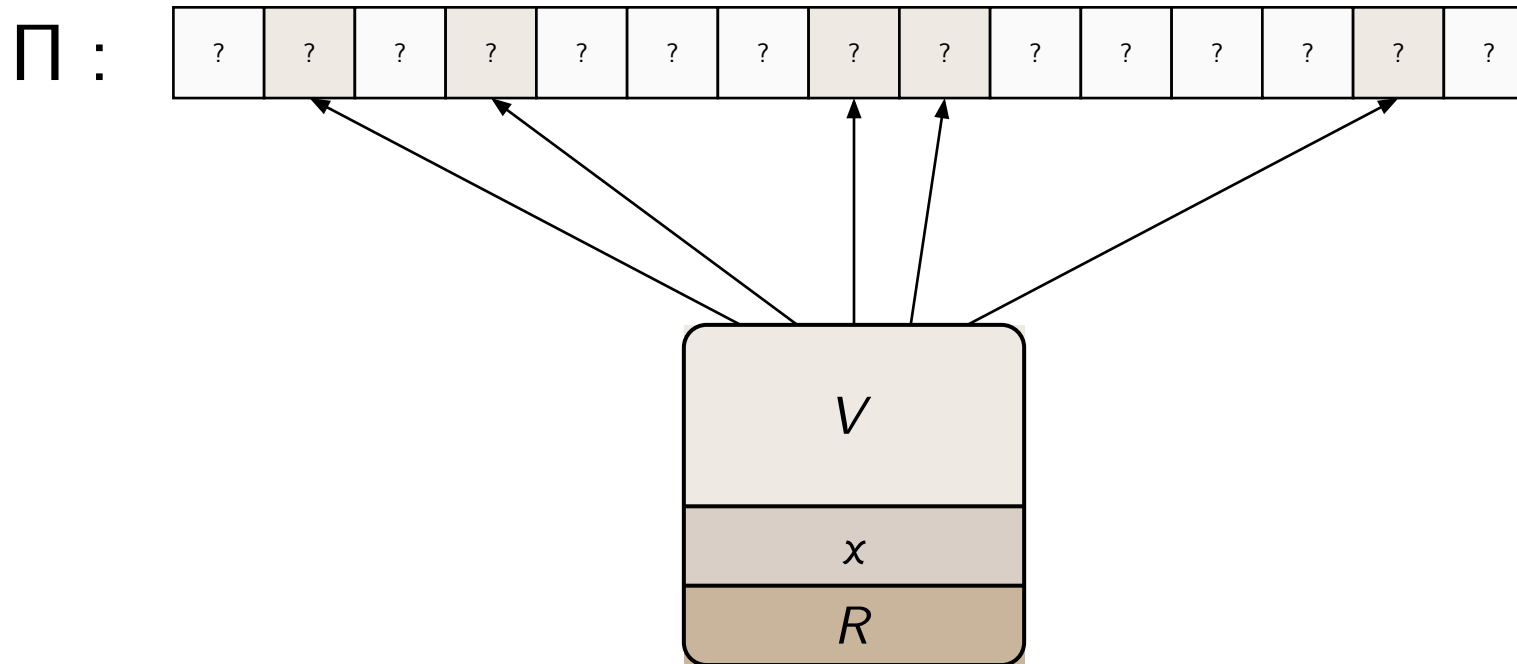


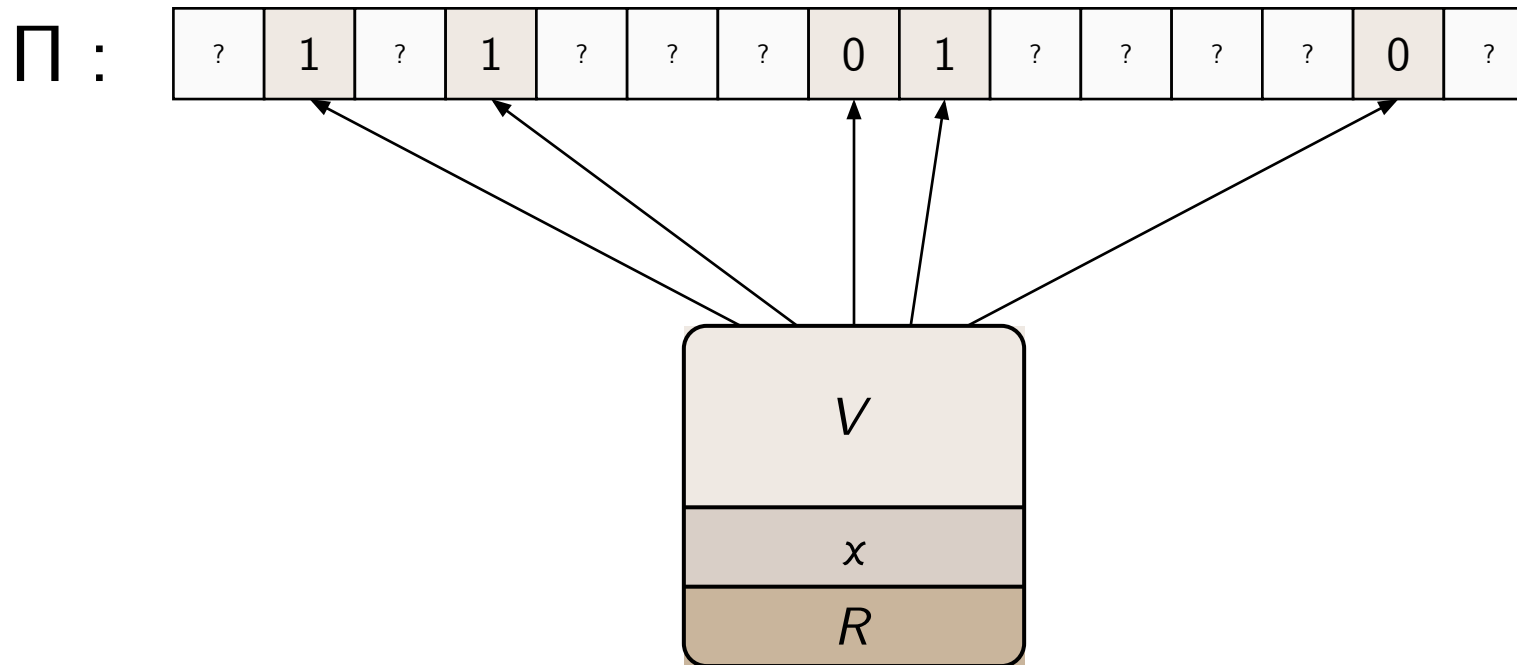
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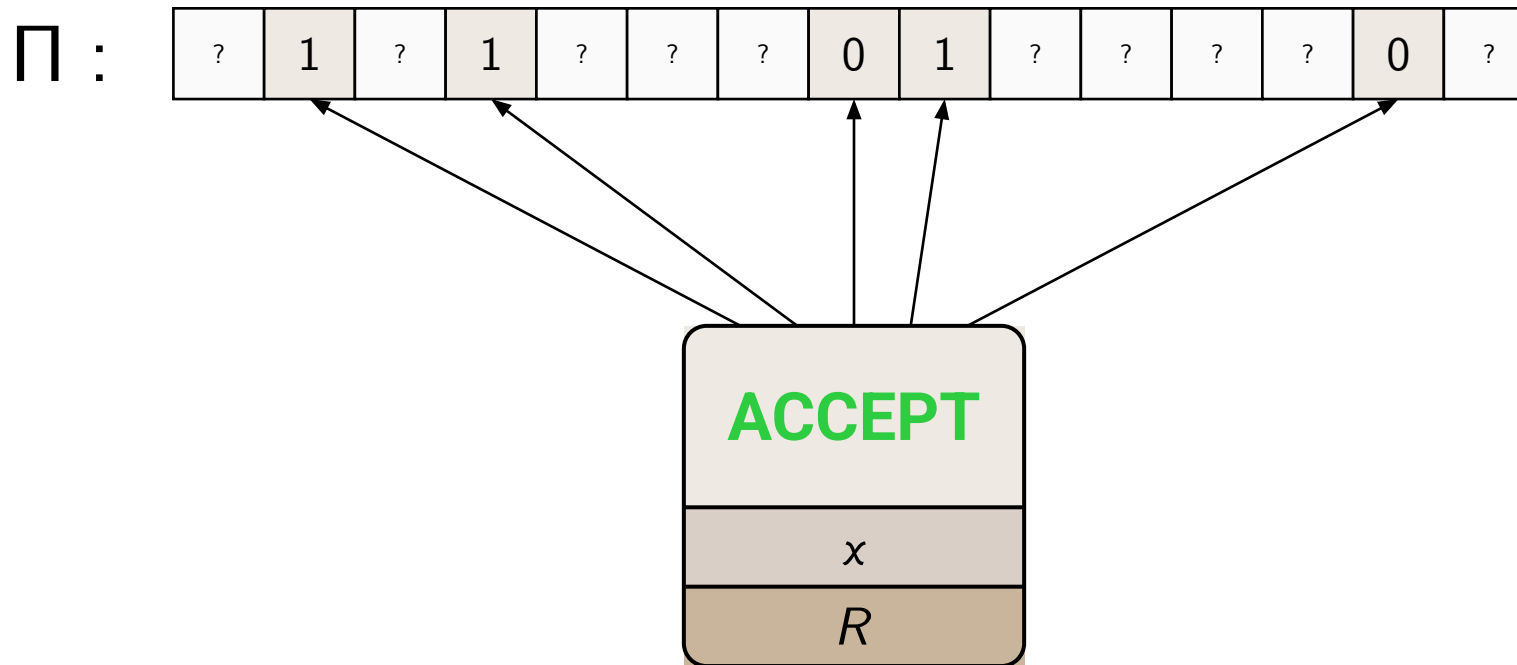
$\Pi :$

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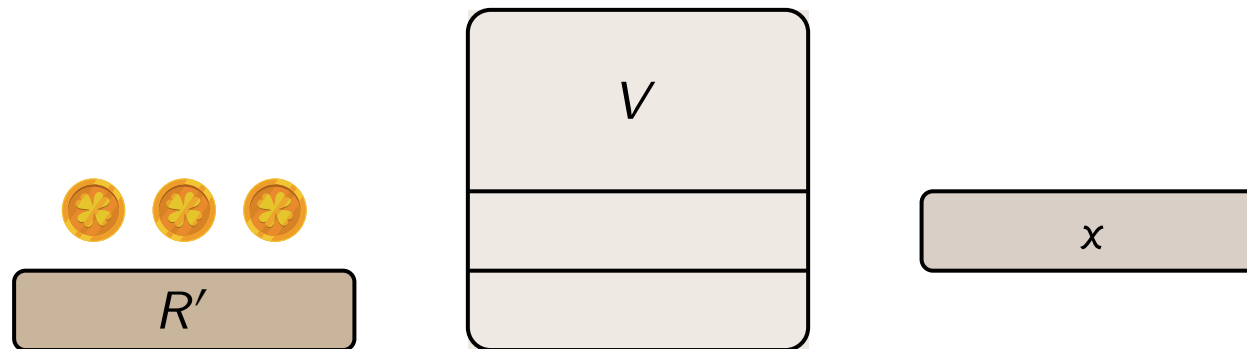




The PCP Theorem

Π :

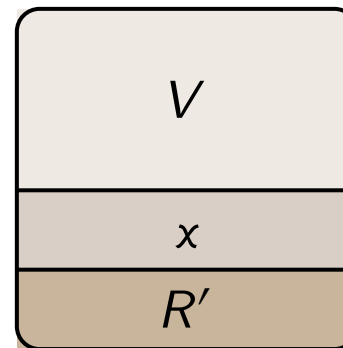
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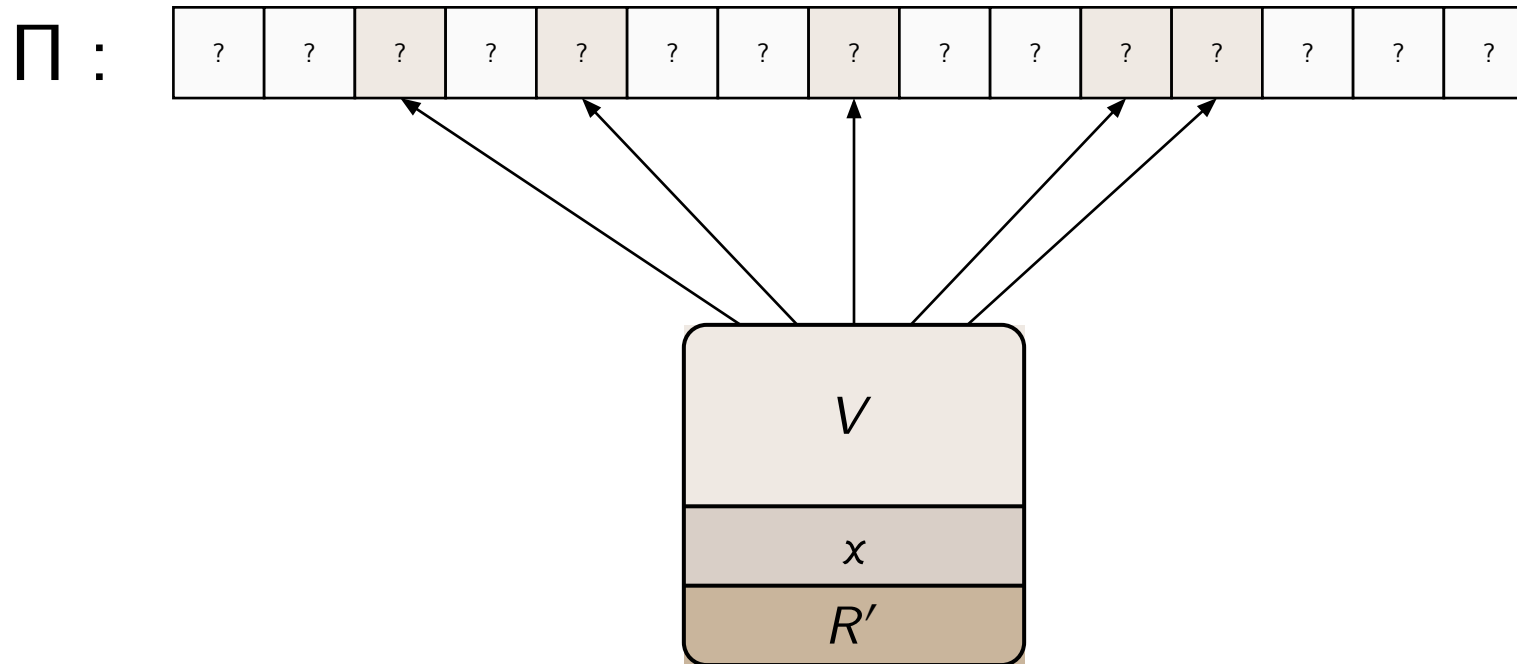


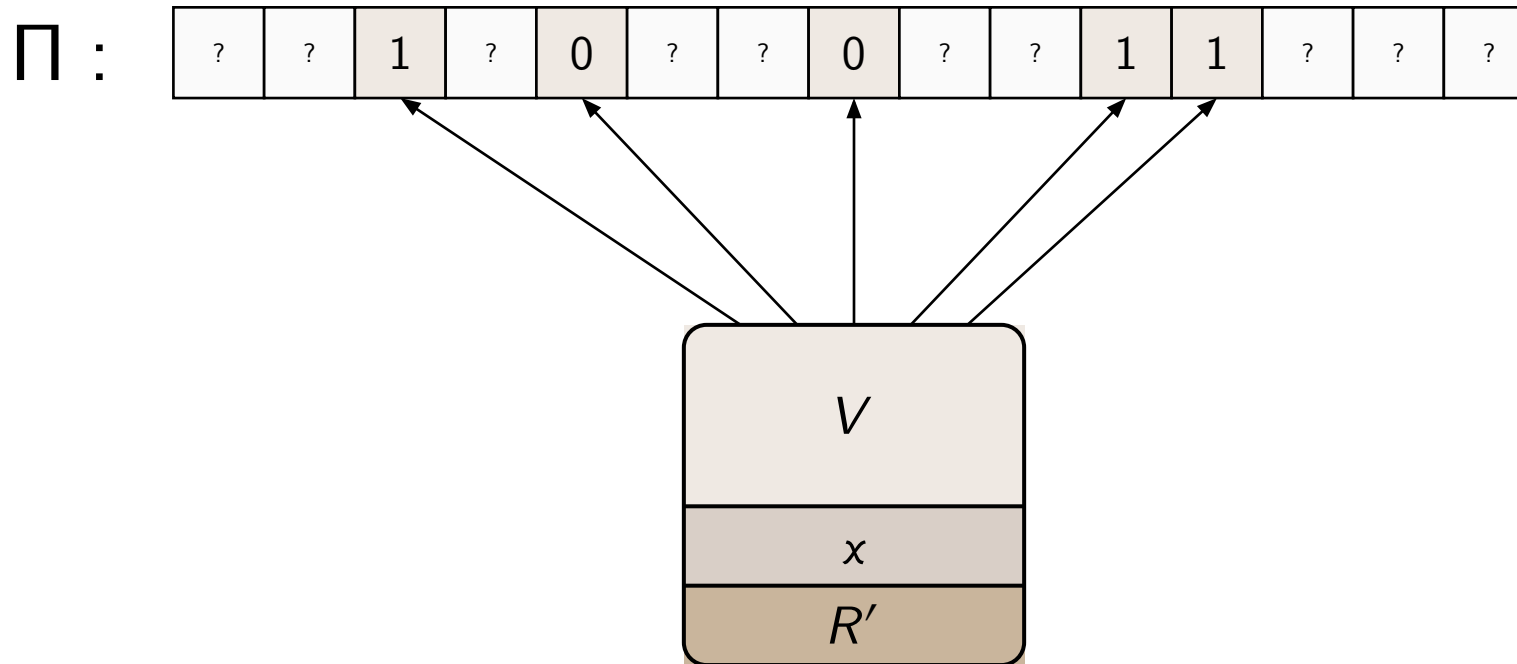
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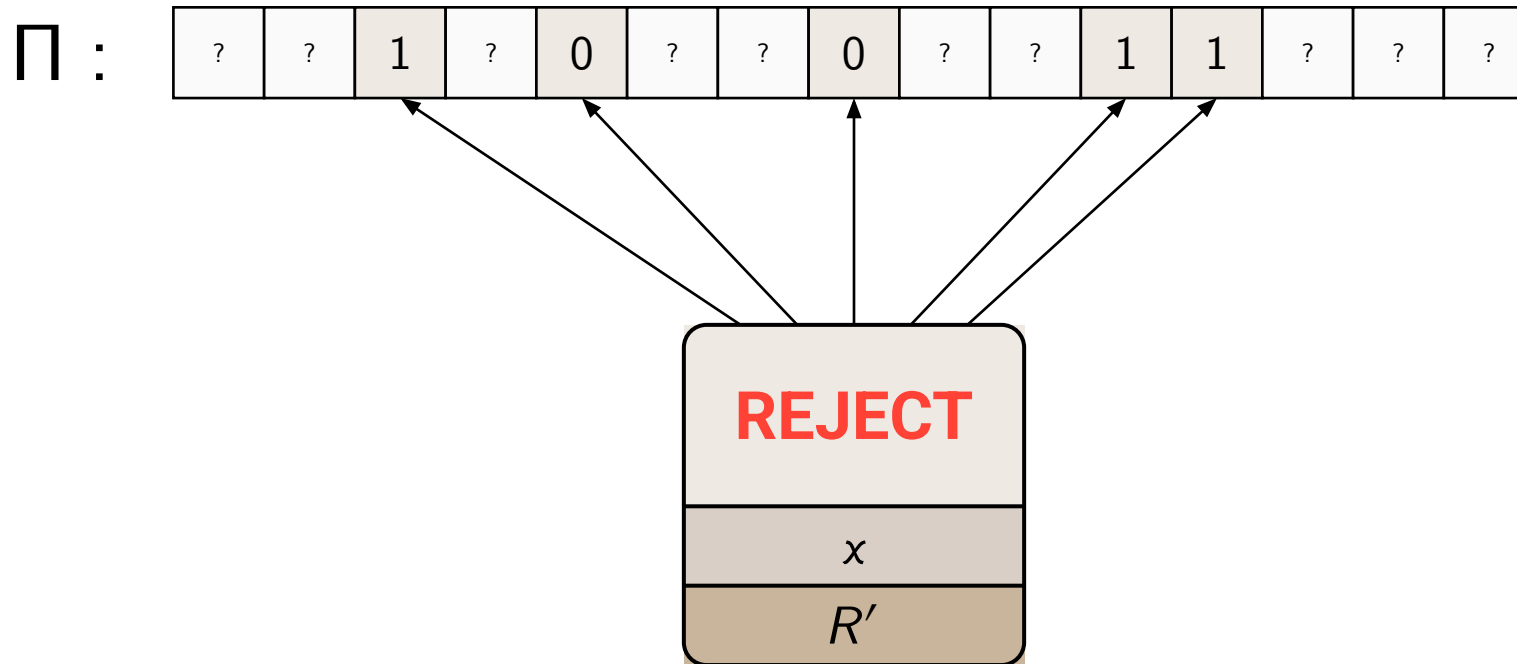
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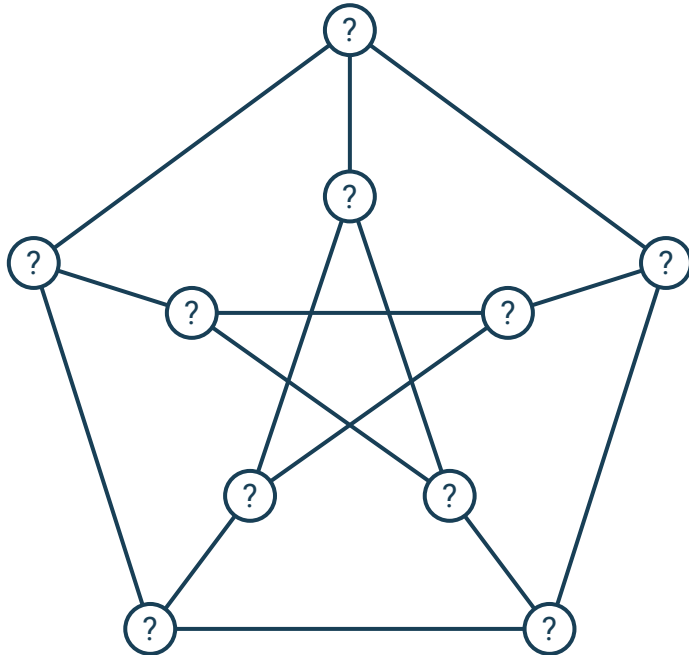






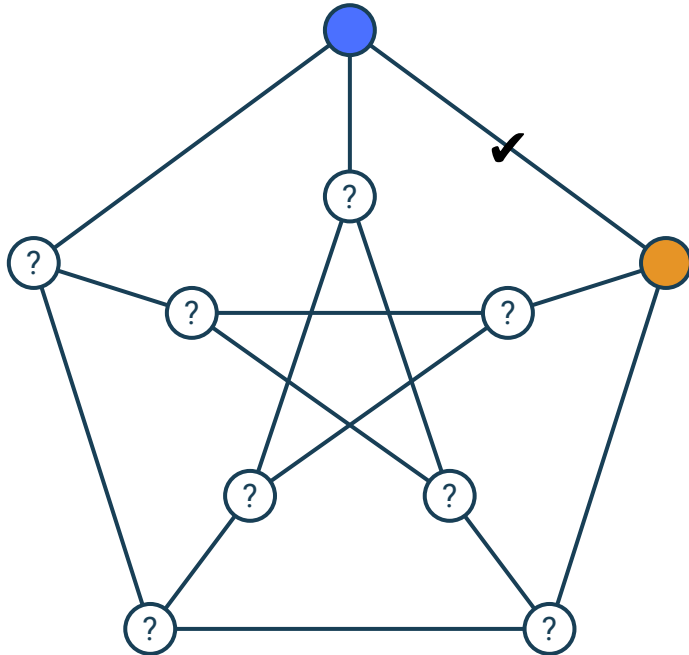


Probabilistic verification of 3-Colorability



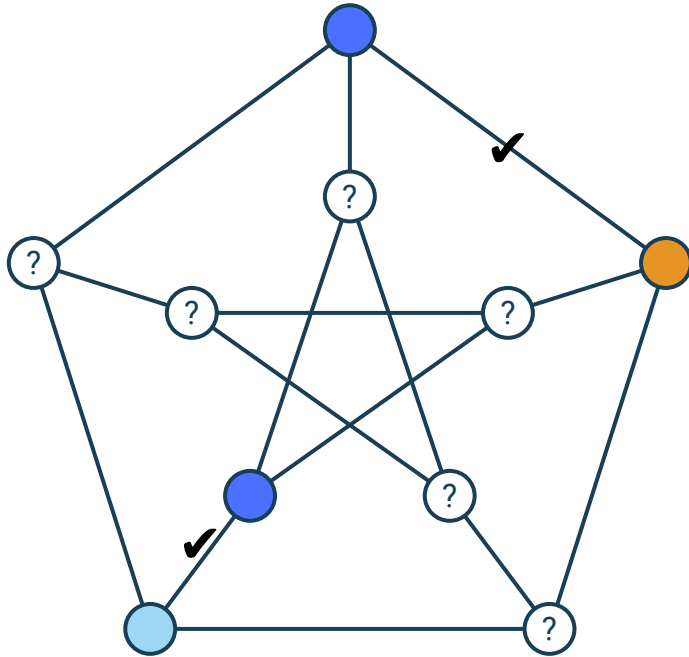
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- V chooses at random (R) a constant number of constraints;
- accept $\iff$ all of them are satisfied.

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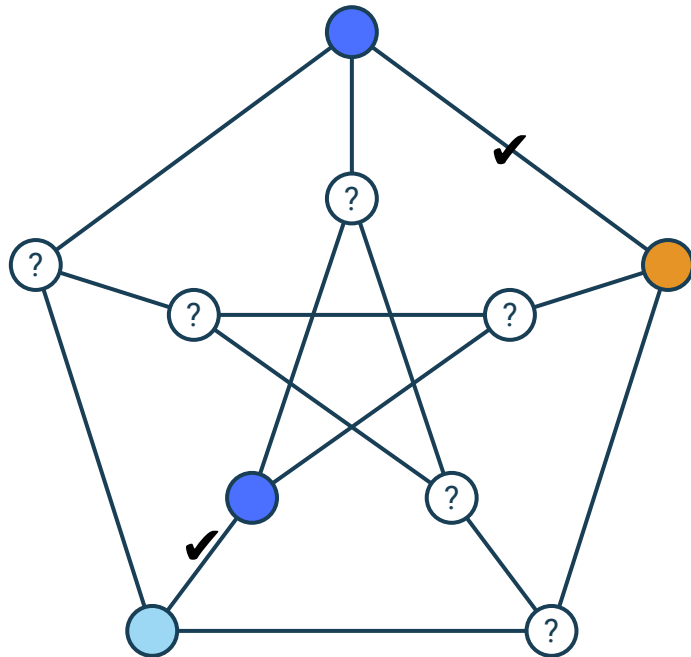
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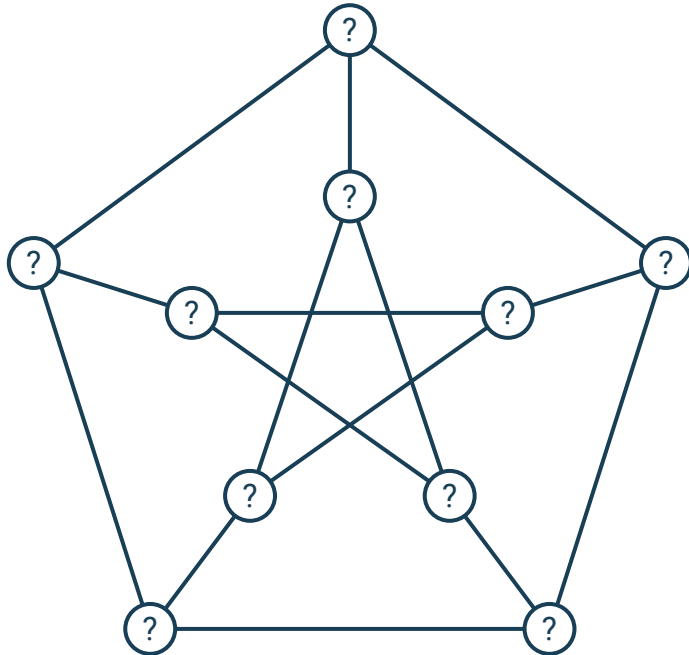
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ACCEPT

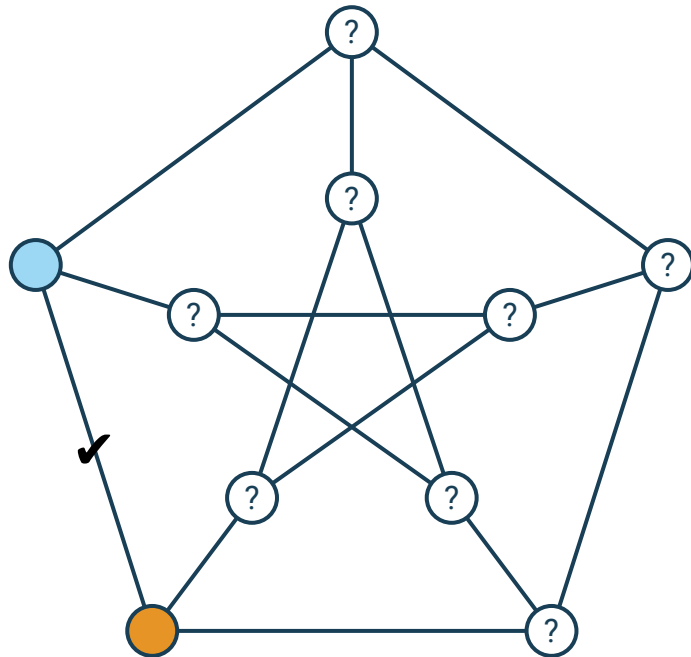
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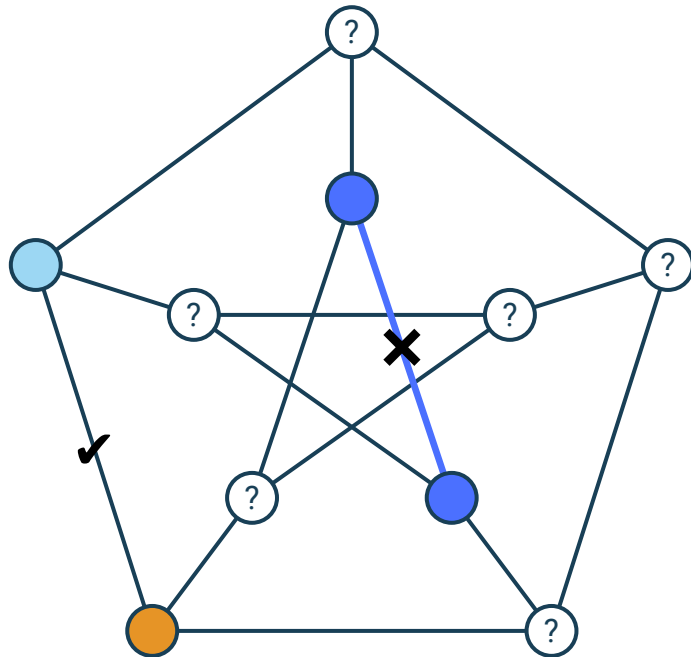
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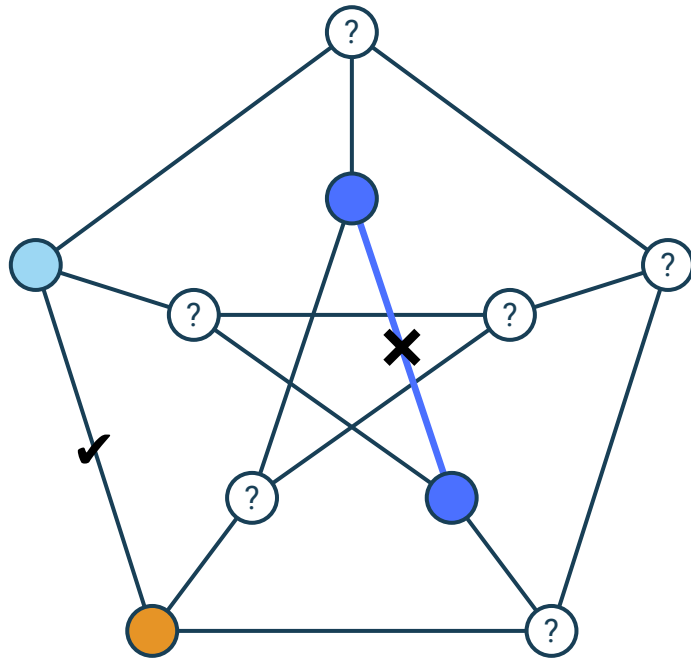
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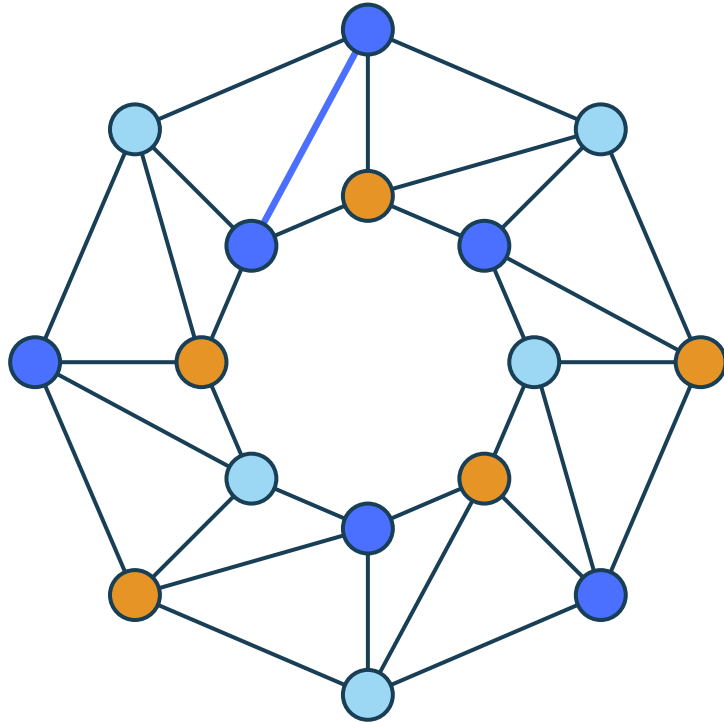
Probabilistic verification of 3-Colorability



REJECT

- Π = coloring;
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- accept $\iff$ all of them are satisfied.

Issue



Colorings could violate only
a fraction $\frac{1}{m}$ of the constraints.



Very unlikely to catch
the unsatisfied edge...

The class $\text{PCP}(O(\log n), O(1))$

A decision problem belongs to the class $\text{PCP}(O(\log n), O(1))$ if there exists a verifier V as above s.t.:

on a YES instance

There is a $\Pi : \mathbb{Z}^+ \rightarrow \{0, 1\}$ s.t.
 $\Pr[V \text{ accepts}] = 1.$

on a NO instance

For any $\Pi : \mathbb{Z}^+ \rightarrow \{0, 1\}$ it
holds $\Pr[V \text{ accepts}] < \frac{1}{2}.$

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Observation: by $(c, q) \mapsto (kc, kq)$, we can make $\Pr[V \text{ is wrong}] < \frac{1}{2^k}.$

So, is 3-Colorability in $PCP(O(\log n), O(1))$?

So, is 3-Colorability in $\text{PCP}(O(\log n), O(1))$?

YES!

So, is 3-Colorability in $\text{PCP}(O(\log n), O(1))$?

YES!

Key

There exists an efficient transformation $G \mapsto G'$ such that

- G 3-colorable $\Rightarrow G'$ 3-colorable;
- G not 3-colorable $\Rightarrow$ every coloring of G' violates at least a constant fraction of constraints.

The PCP Theorem

Arora, Safra (1992); Arora, Lund, Motwani, Sudan, Szegedy (1992); Dinur (2006)

$$\text{NP} = \text{PCP}(O(\log n), O(1)).$$

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Max-Clique

Definition (Clique)

A *clique* in a graph $G(V, E)$ is a complete subgraph of G ,
i.e. a subset of pairwise adjacent nodes.

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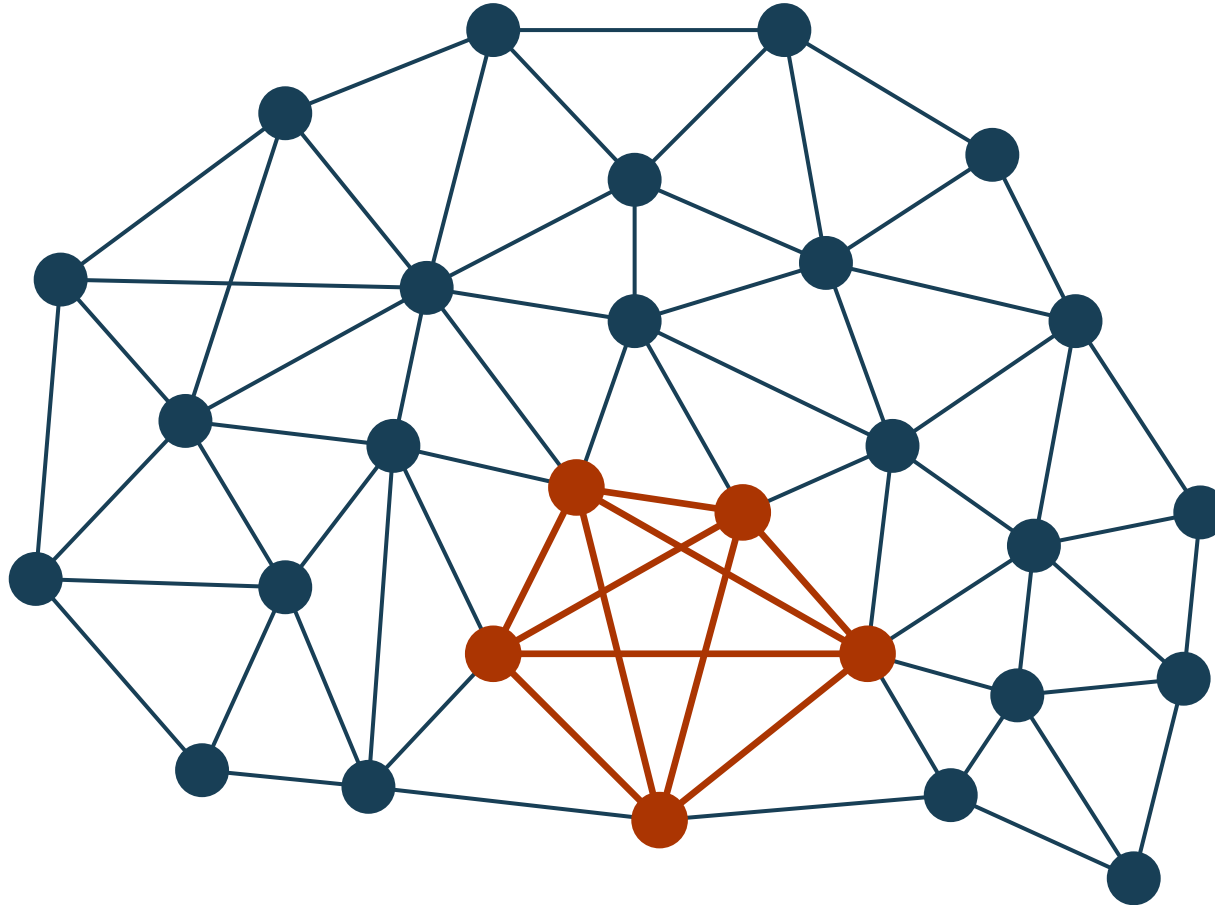
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i.e. a subset of pairwise adjacent nodes.

Problem (Max-Clique)

Given G , find a clique of maximum cardinality.

Let $\omega(G)$ be this cardinality.

Max-Clique



Max-Clique is Hard

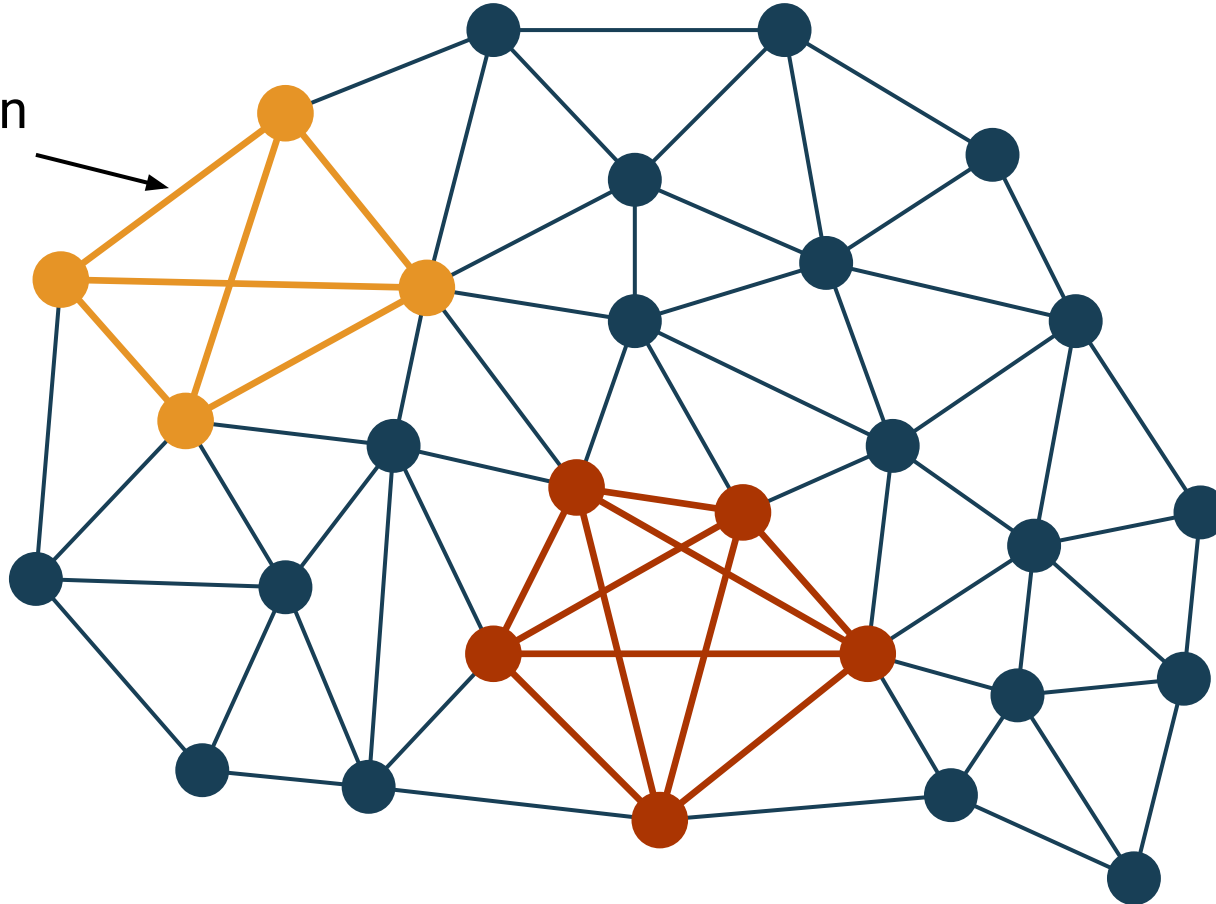
3. CLIQUE
INPUT: graph G , positive integer k
PROPERTY: G has a set of k mutually adjacent nodes.

Karp (1972)

If there is an efficient algorithm for Max-Clique, then $P = NP$.

Can we approximate it?

$\frac{4}{5}$ -approximation



Hardness of approximating Max-Clique

Feige, Goldwasser, Lovász, Safra, Szegedy (1991)

If there is an efficient ε -approximation of Max-Clique for some $\varepsilon > 0$, then $P = NP$.

Proof

We will show how to use an efficient $\bar{\varepsilon}$ -approximation of Max-Clique to decide 3-Colorability efficiently. From now on, $\bar{\varepsilon} = \frac{1}{2}$.

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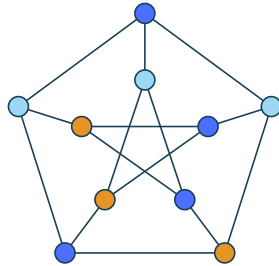
Fundamental reduction

There is an efficient transformation of graphs $G \mapsto G'$ s.t.:

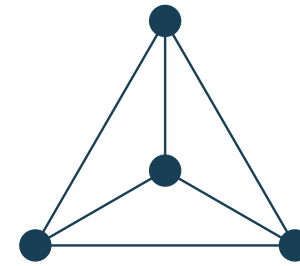
- G is 3-colorable $\Rightarrow \omega(G') = M$;
- G is not 3-colorable $\Rightarrow \omega(G') < M/2$.

Decision procedure

Let K be an $\frac{1}{2}$ -approximation of $\omega(G')$.



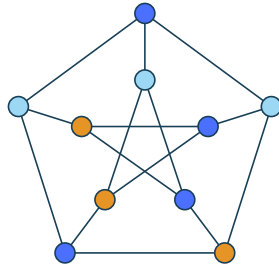
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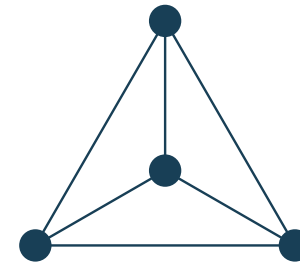
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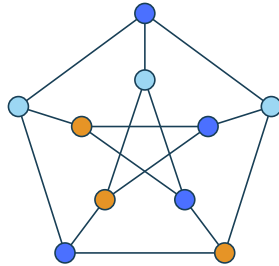
$$|K| \geq M/2$$



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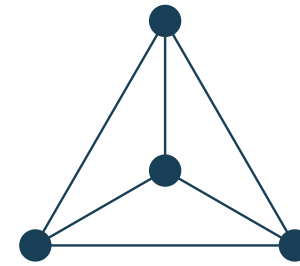
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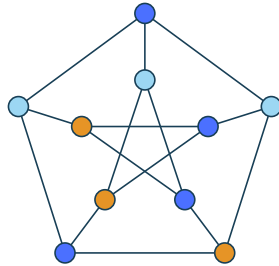


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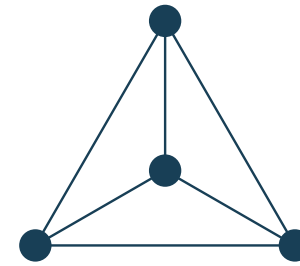
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$$|K| < M/2$$

$$G \text{ is 3-colorable} \Leftrightarrow |K| \geq M/2.$$

Building G'

Fix a PCP-verifier for 3-Colorability, reading:

- an input graph G ;
- $c \log n$ random bits;
- q bits from the proof.

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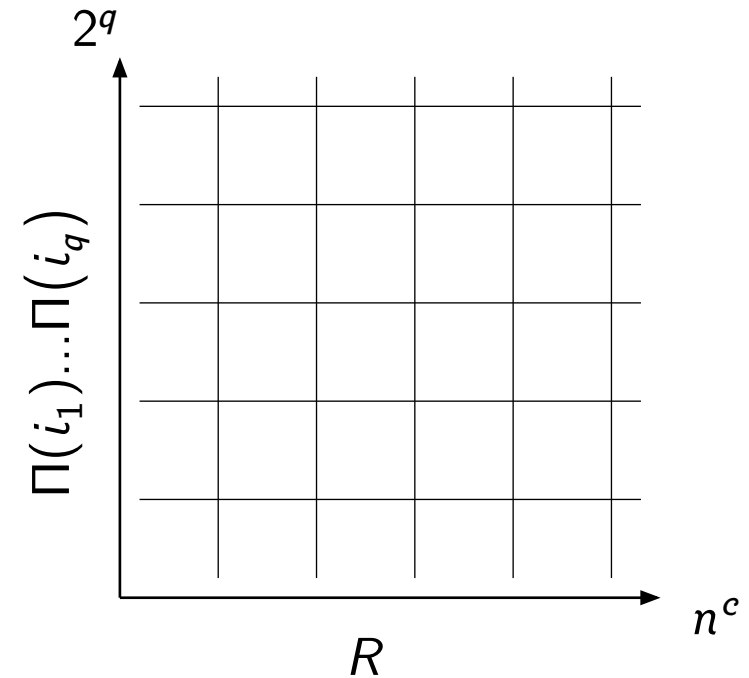
Reminder

Fixed G , the verifier accepts or rejects based solely on R and $\Pi(i_1), \dots, \Pi(i_q)$.

Nodes of G'

The information $\langle R; \Pi(i_1) \dots \Pi(i_q) \rangle$ is an *accepting transcript* if:

- on the random string R , the verifier reads the bits at positions $i_1, \dots, i_q$ and
- reading the bits $\Pi(i_1), \dots, \Pi(i_q)$ causes the verifier to accept.

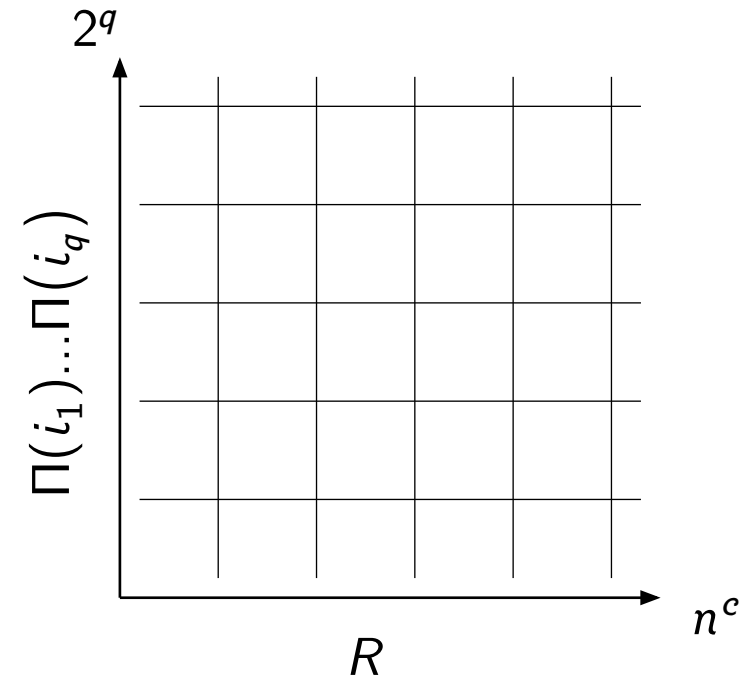


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Nodes of $G' = \{ \langle R; \Pi(i_1) \dots \Pi(i_q) \rangle : V \text{ accepts} \}$

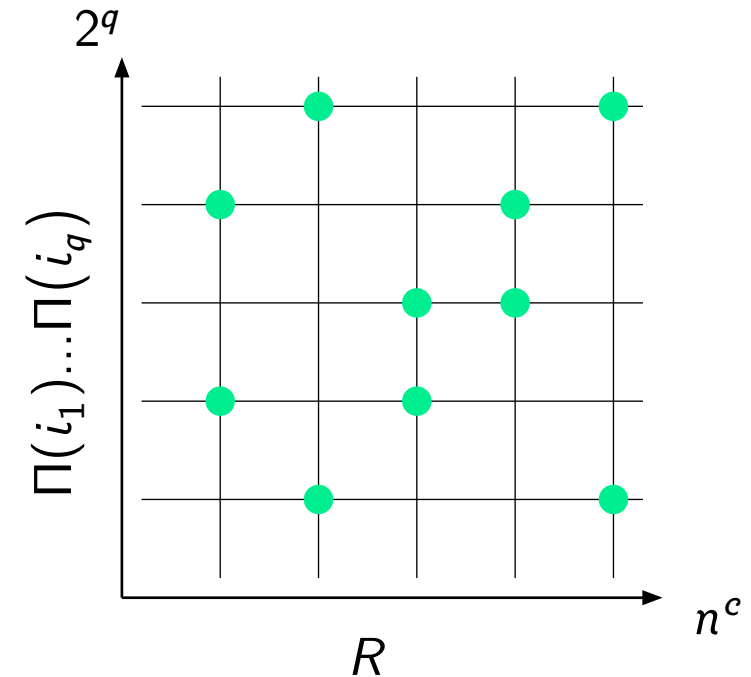


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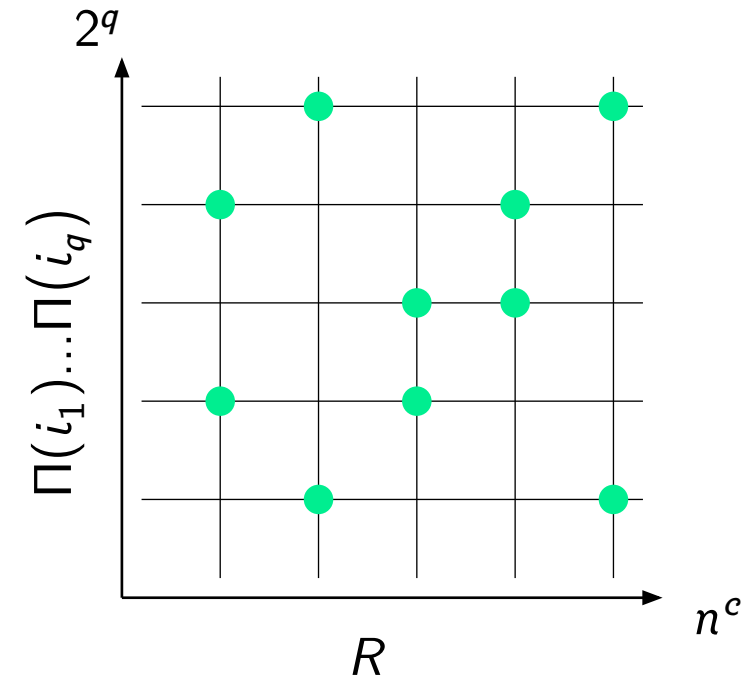
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Edges of G'

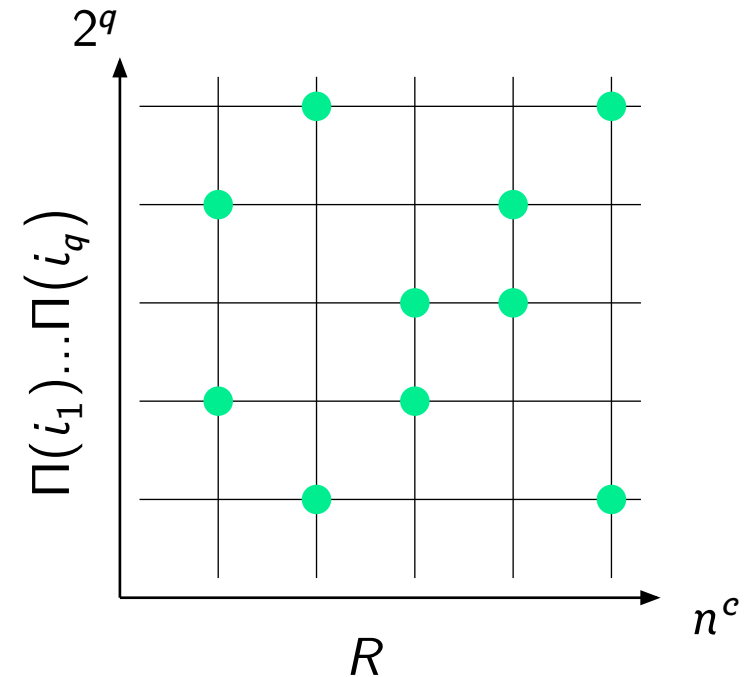
Two accepting transcripts $\langle R_1; \Pi(i_1) \dots \Pi(i_q) \rangle$ and $\langle R_2; \Pi(j_1) \dots \Pi(j_q) \rangle$ are *consistent* if they do not disagree on any bit of the proof, i.e. $i_a = j_b$ implies $\Pi(i_a) = \Pi(j_b)$.



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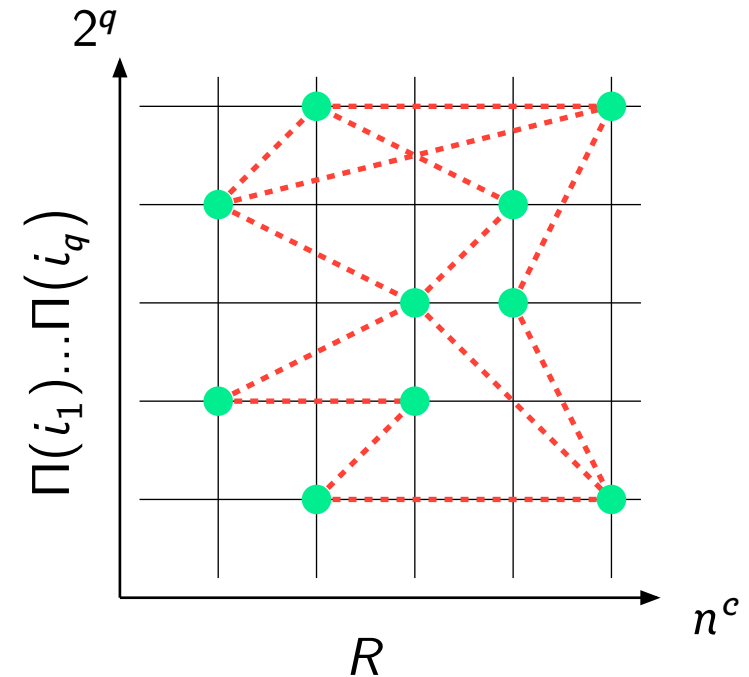
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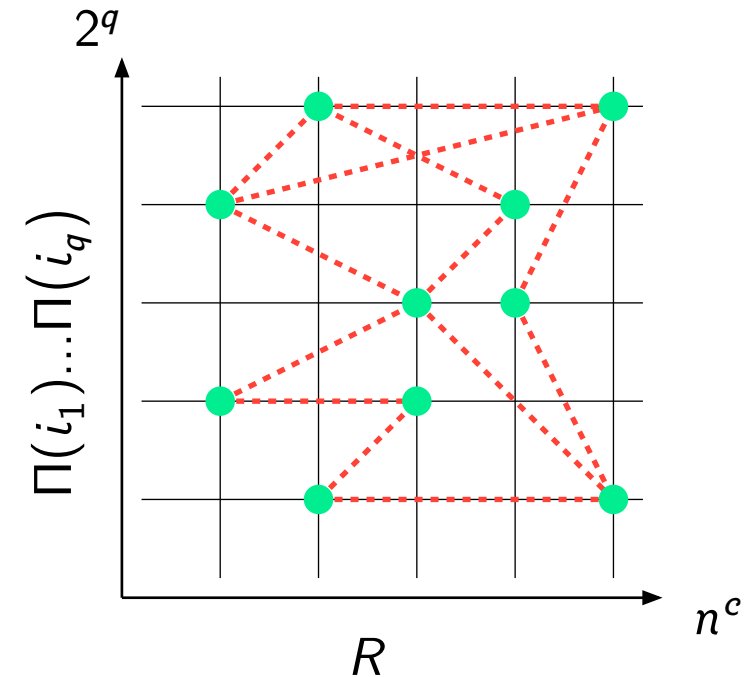
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Cliques $\Leftrightarrow$ Proofs

Cliques K in G' correspond to partial proofs $\tilde{\Pi}_K : S \subseteq \mathbb{Z}^+ \rightarrow \{0, 1\}$, where for every Π_K extending $\tilde{\Pi}_K$

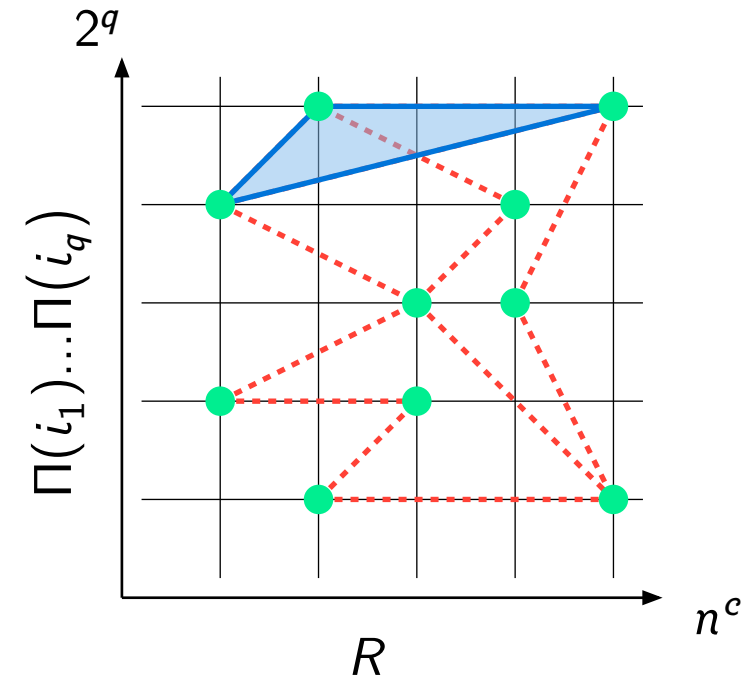
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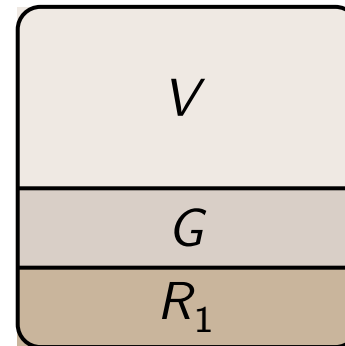
$\Pi :$

?	?	?	?	?	?	?	?	?	?	?	?	?	?	?
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

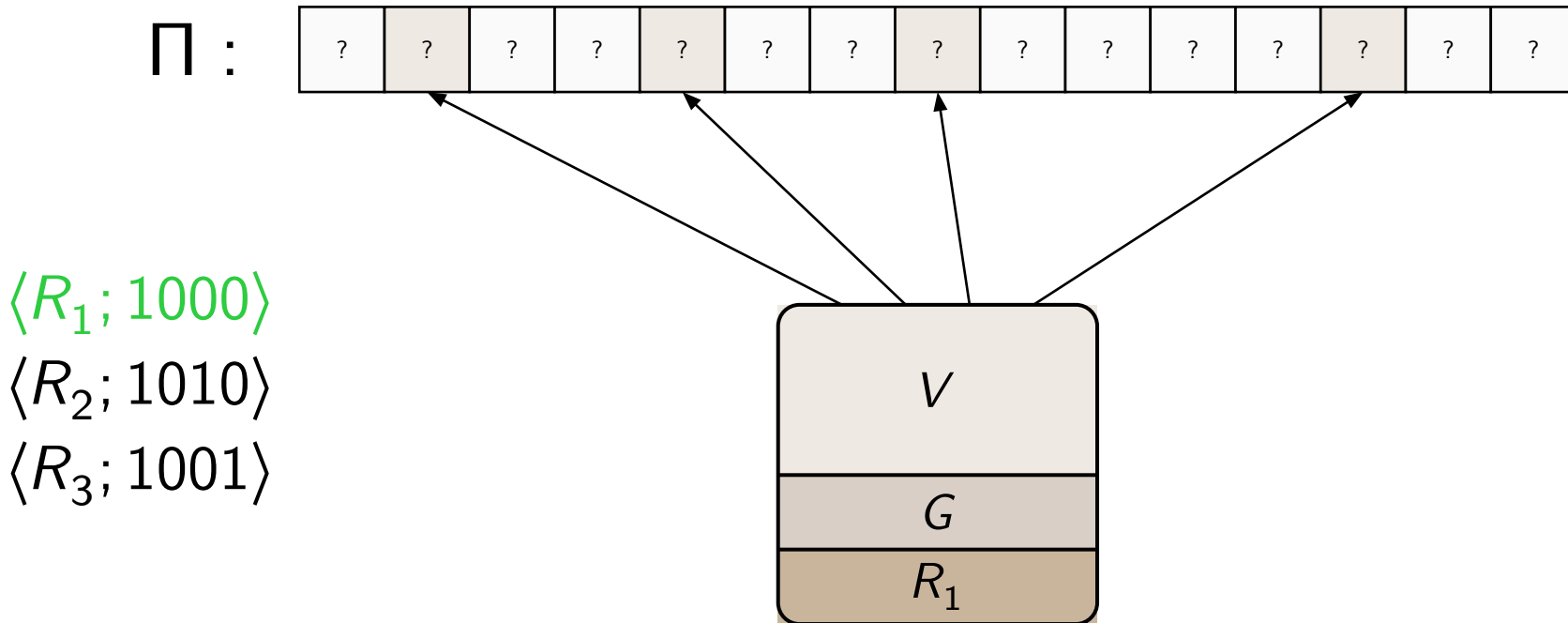
$\langle R_1; 1000 \rangle$

$\langle R_2; 1010 \rangle$

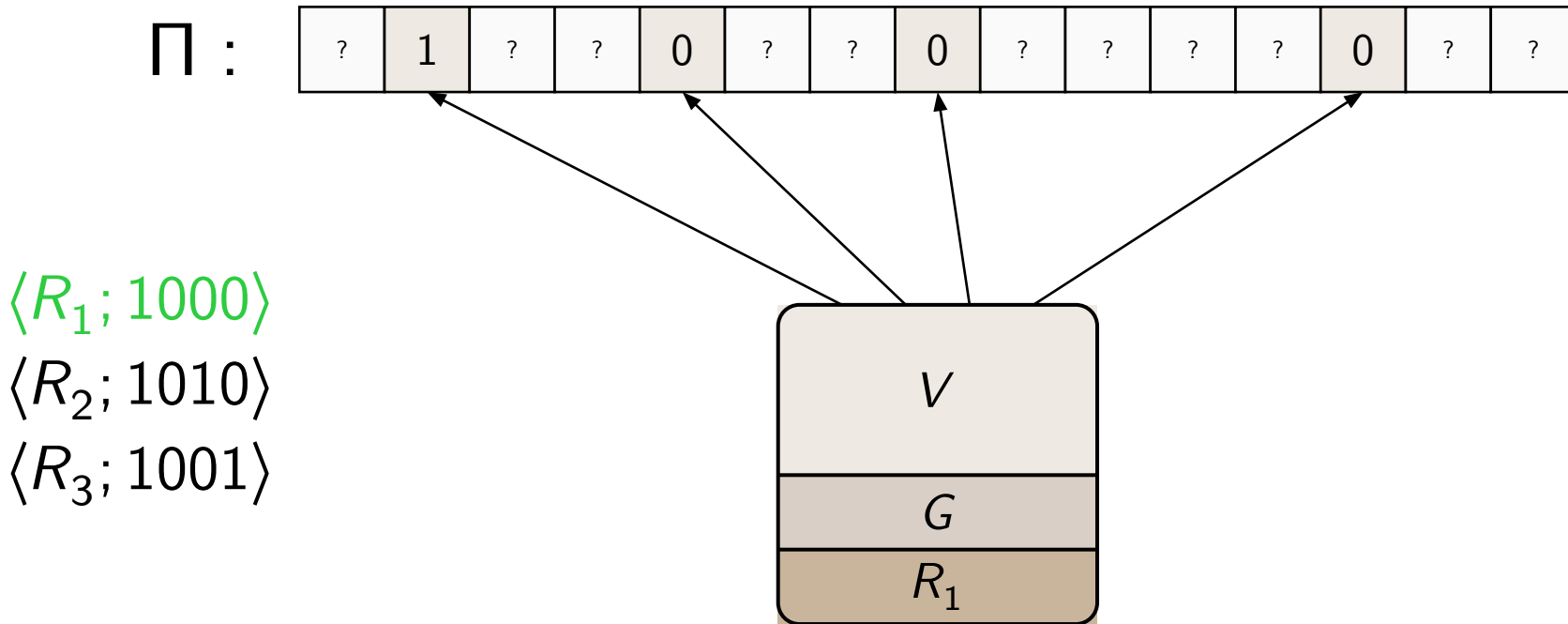
$\langle R_3; 1001 \rangle$



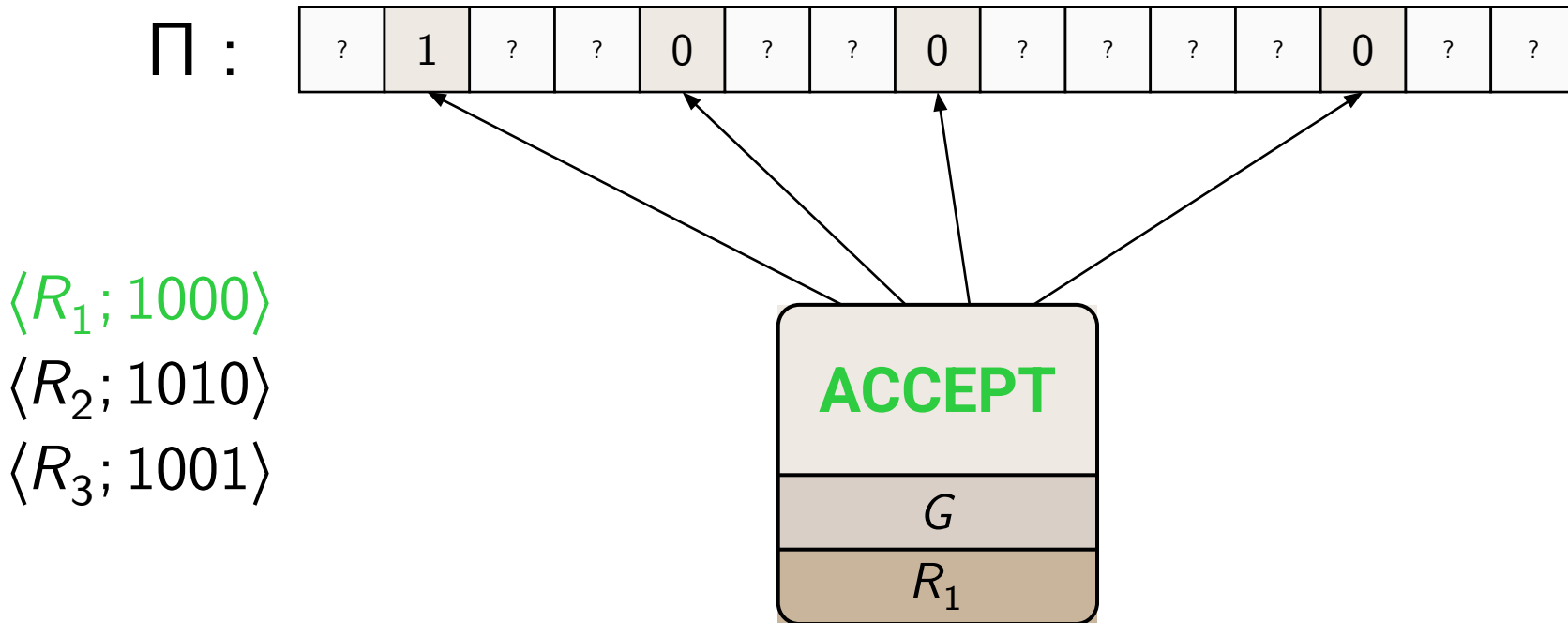
Cliques $\Leftrightarrow$ Proofs



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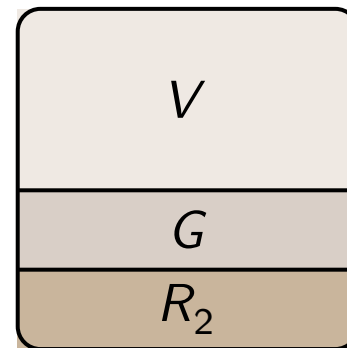
$\Pi :$

?	1	?	?	0	?	?	0	?	?	?	?	0	?	?
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

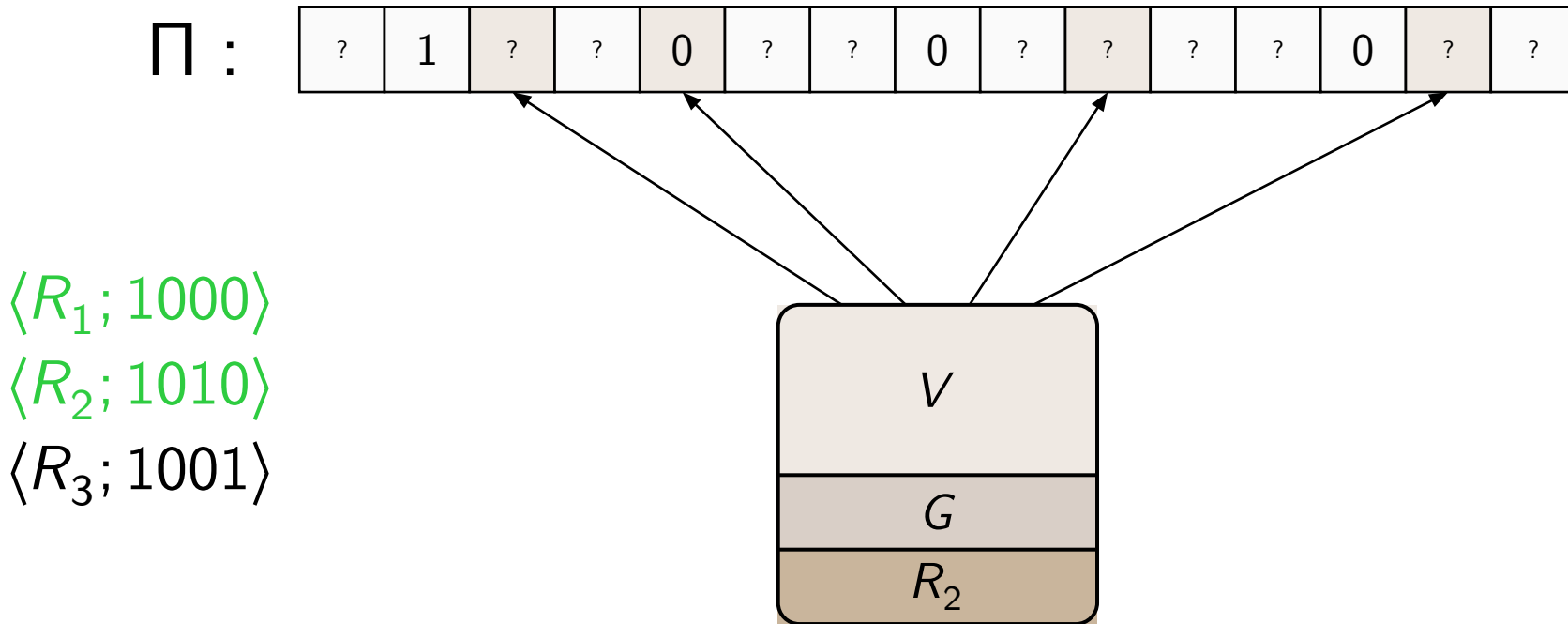
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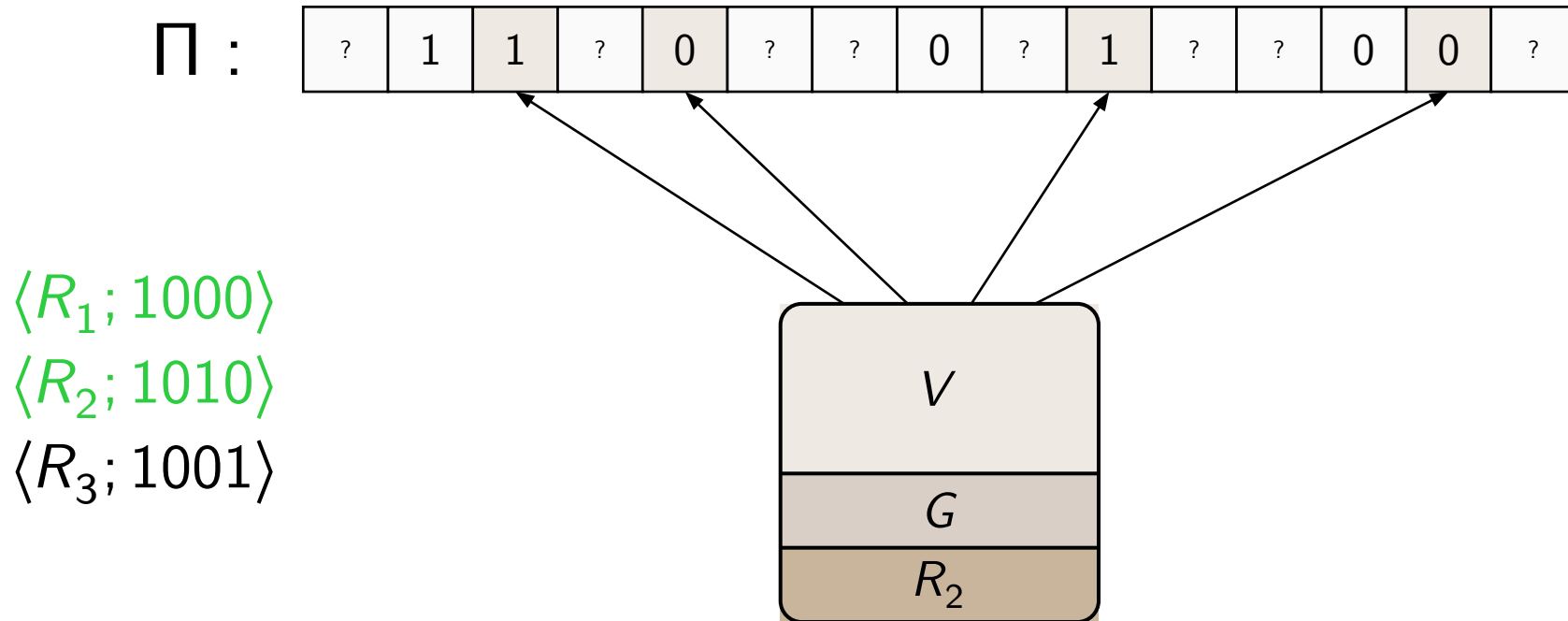
$\langle R_3; 1001 \rangle$



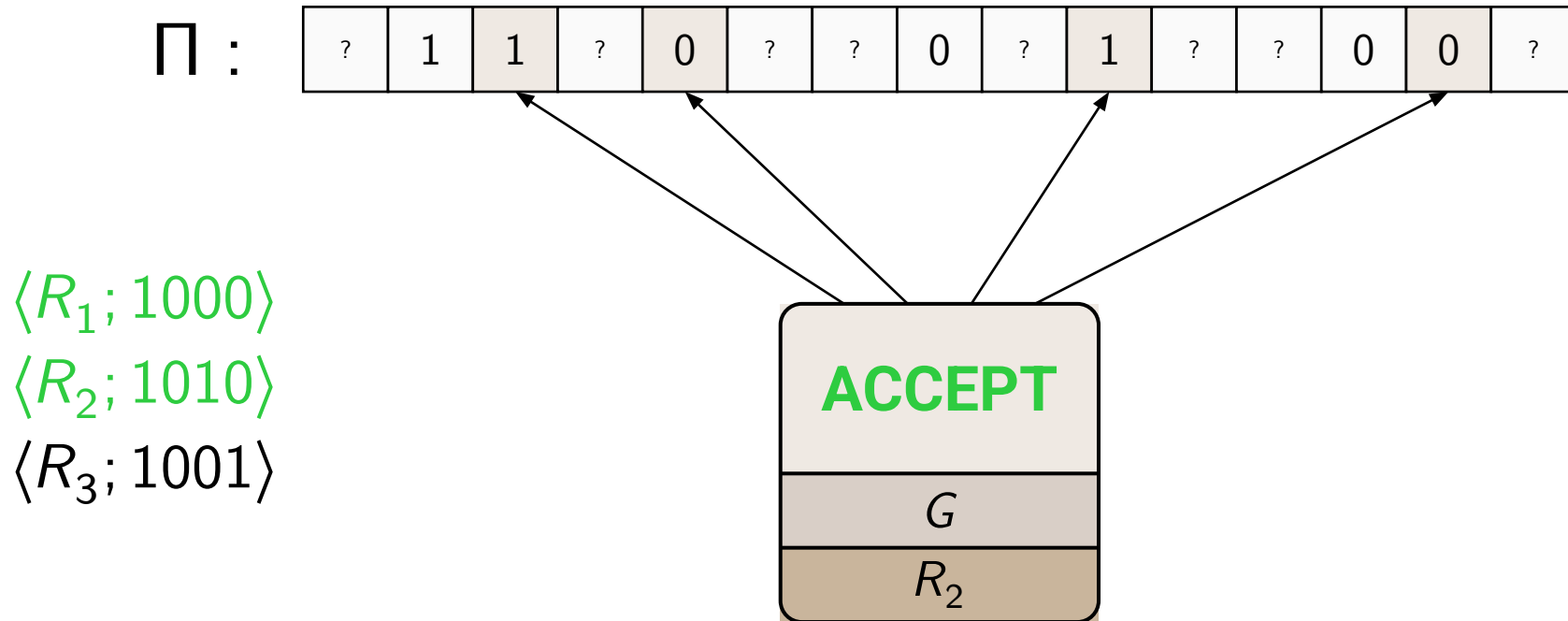
Cliques $\Leftrightarrow$ Proofs



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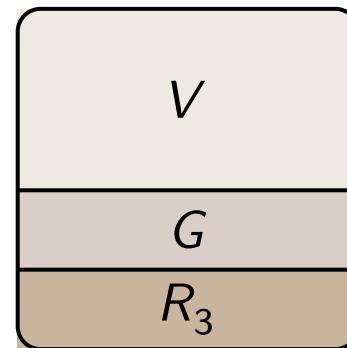
$\Pi :$

?	1	1	?	0	?	?	0	?	1	?	?	0	0	?
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

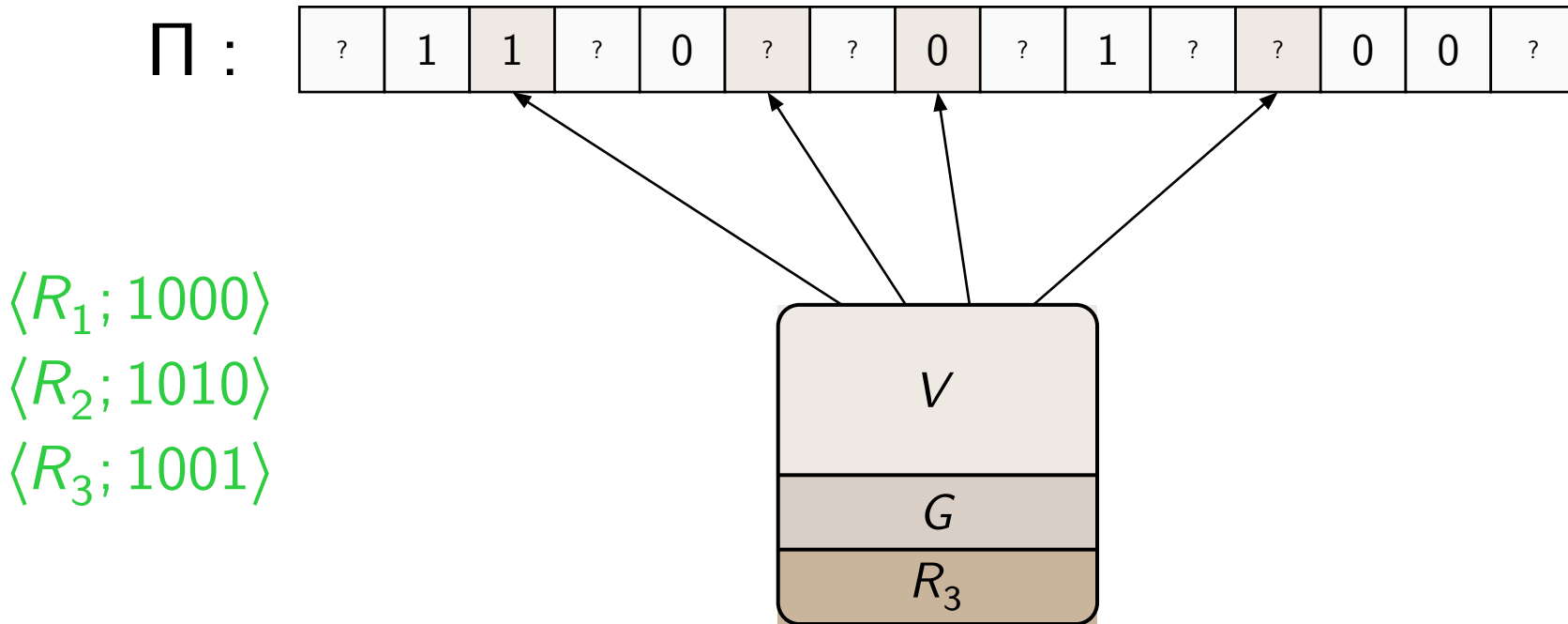
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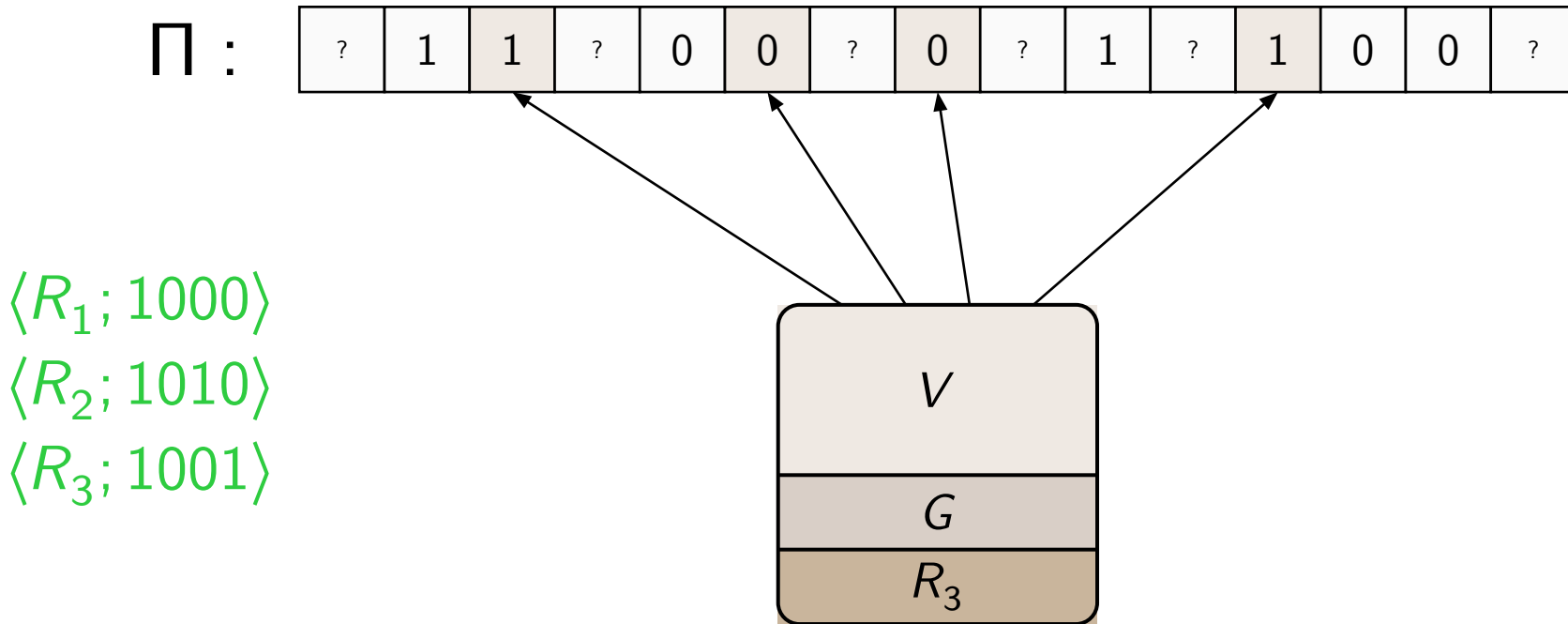
$\langle R_3; 1001 \rangle$



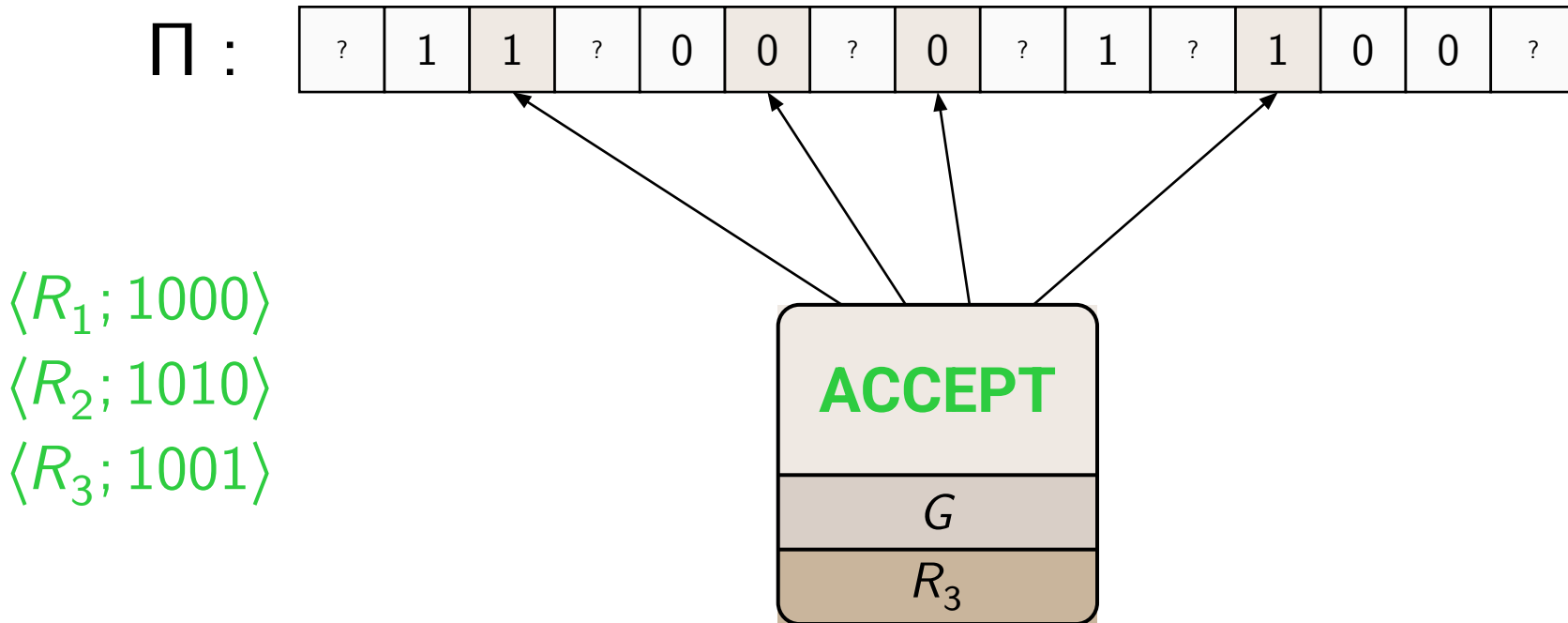
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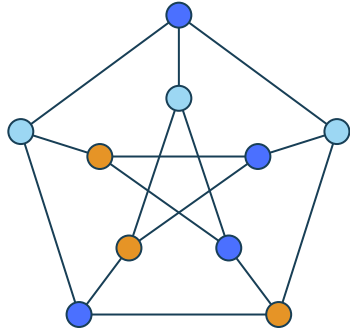
Cliques $\Leftrightarrow$ Proofs



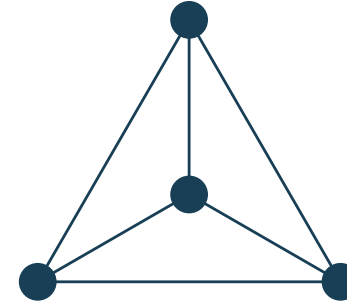
Cliques $\Leftrightarrow$ Proofs



Putting all pieces together

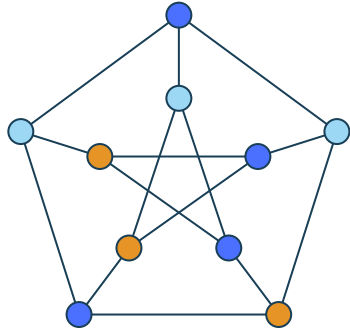


There is a
valid proof Π



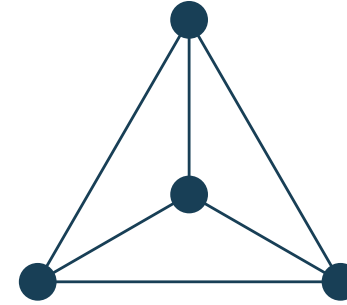
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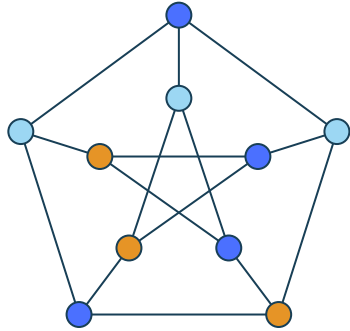
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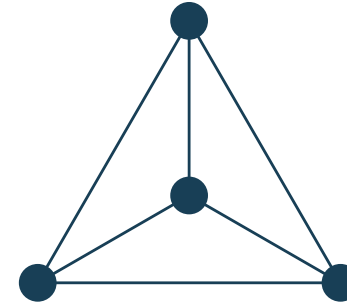
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Even stronger inapproximability results

Our proof...	...with a pinch of expander graphs	A stronger PCP + ...
ε for any $\varepsilon > 0$	$\frac{1}{n^c}$ for some $c > 0$	$\frac{1}{n^{1-\varepsilon}}$ for any $\varepsilon > 0$

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The PCP Theorem changed an era in approximation theory.

1. P vs NP
2. The PCP Theorem
3. Inapproximability of Max-Clique
4. **Landscape**

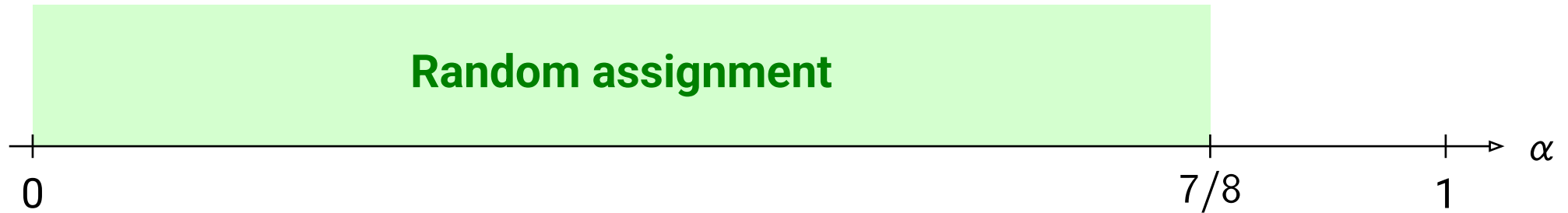
Max-E3Sat

Max-E3Sat

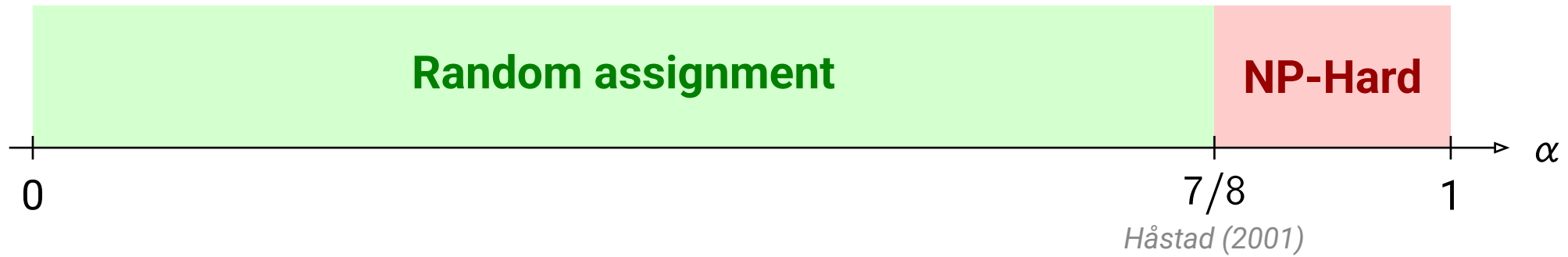
Given a boolean 3-CNF formula, find a truth assignment maximizing the number of satisfied clauses.

$$(x_1 \vee x_2 \vee \neg x_4) \wedge (x_3 \vee \neg x_1 \vee x_5) \wedge \dots \wedge (\neg x_{256} \vee \neg x_{42} \vee x_8)$$

Max-E₃Sat



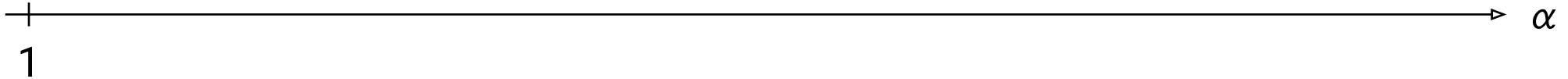
Max-E₃Sat



Set Cover

Problem (Set-Cover)

Given a collection of subsets $S_1, \dots, S_m$ of a set U such that $\bigcup_i S_i = U$, find a smallest set $I \subseteq \{1, \dots, m\}$ such that $\bigcup_{i \in I} S_i = U$.



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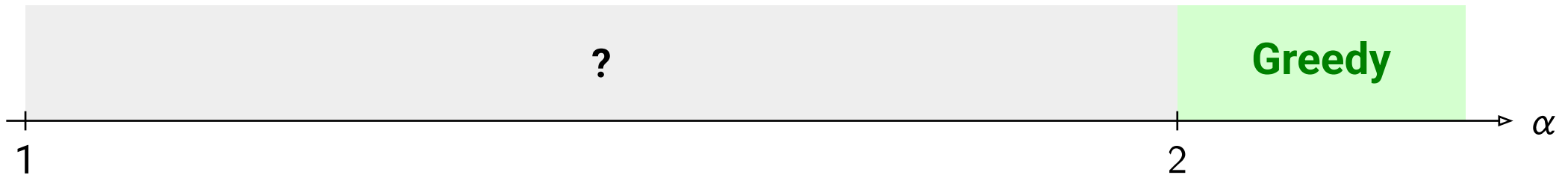


Feige (1998)
Dinur & Steurer (2014)

Vertex Cover

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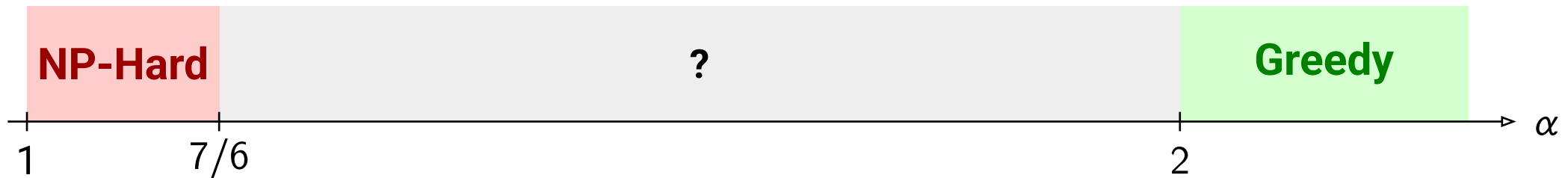
Given a graph $G(V, E)$, find a smallest set of nodes $C \subseteq V$, such that for each $\{u, v\} \in E$, either $u \in C$ or $v \in C$.



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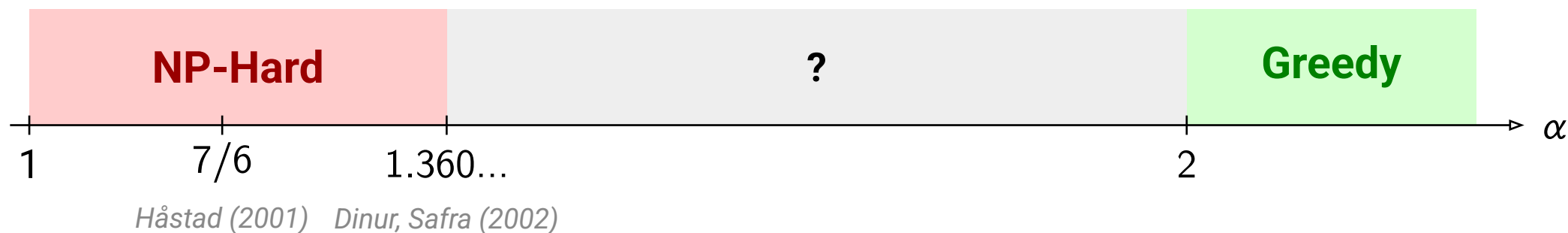


Håstad (2001)

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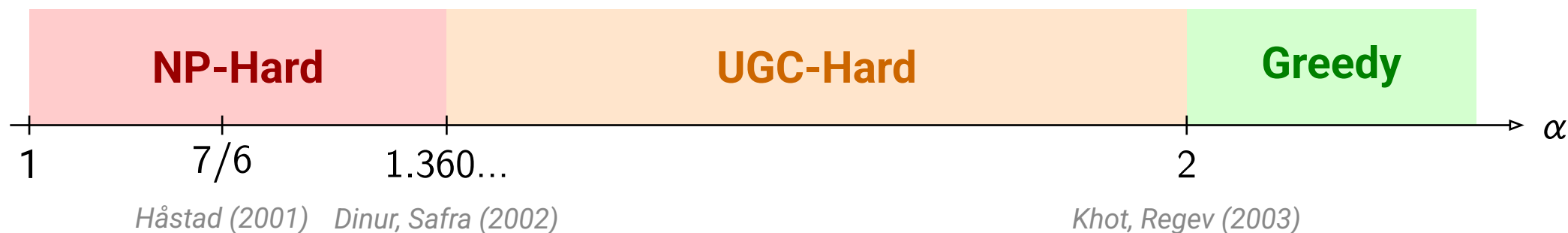
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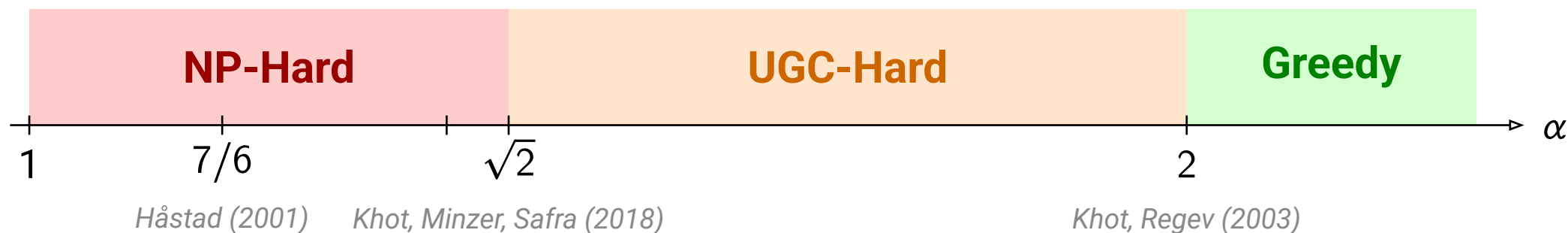
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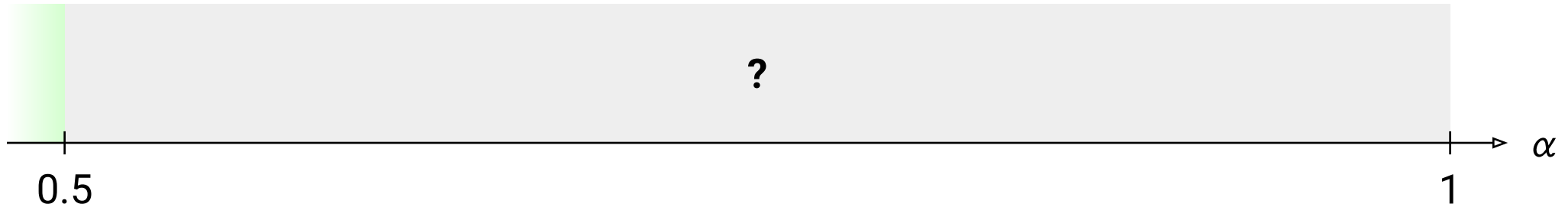
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(Partial) history of the PCP Theorem

- (1971) Cook – *The complexity of theorem-proving procedures*
- (1972) Karp – *Reducibility among combinatorial problems*
- (1973) Levin – *Universal Sequential Search Problems*
- (1991) Feige, Goldwasser, Lovász, Safra, Szegedy – *Interactive Proofs and the Hardness of Approximating Cliques*
- (1992) Arora, Safra – *Probabilistic Checking of Proofs: A New Characterization of NP*
- (1992) Arora, Lund, Motwani, Sudan, Szegedy – *Proof Verification and the Hardness of Approximation Problems*
- (2001) Håstad – *Some Optimal Inapproximability Results*
- (2006) Dinur – *The PCP Theorem by Gap Amplification*

Some more bibliography

- (1989) Impagliazzo, Zuckerman – *How to Recycle Random Bits*
- (1995) Buss – *On Gödel's theorems on lengths of proofs II: Lower bounds for recognizing k symbol provability*
- (2004) Trevisan – *Inapproximability of Combinatorial Optimization Problems*
- (2006) Radhakrishnan, Sudan – *On Dinur's Proof of the PCP Theorem*
- (2012) Arora, Barak – *Computational Complexity: A Modern Approach*



SCUOLA
NORMALE
SUPERIORE

Time for your questions!

More topics we could talk about

What expanders
have to do with
the PCP Theorem

A stronger PCP and
optimal inapproximability
results

Why several problems do
not have a PTAS,
unless $P = NP$

Zero-knowledge proofs

Inapproximability results
for other optimization
problems